

The S-Plane Project, from the Beginning

Open Fronthaul timing security in O-RAN, told as one continuous story: what the problem is, how time travels, what we built, what the deck says slide by slide, and what the numbers really mean.

WORKLET	AI-Native Self-Healing O-RAN Network using a Digital Twin
SUB-SCOPE	Open Fronthaul S-Plane Timing Security
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SCOPE OF THIS GUIDE	Software work only. Every number is taken from the corrected 168-run campaign files, the v8 deck and its verification records.

How to read this. Part I is the story of the project (about 25 minutes). Part II walks through all 42 slides of the v8 deck with what each one says and what to say. Part III is the results, explained down to the arithmetic, then the questions you are most likely to be asked. If you only have ten minutes, read the one-page summary that follows the contents.

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The whole project on one page

The setting. A 5G O-RAN base station is split into separate boxes. The radio unit (O-RU) sits at the antenna and is fed by the distributed unit (O-DU) over an Ethernet link called the **Open Fronthaul**. The radio units must agree on time to within about a hundred nanoseconds. Time is not sent on a special wire. It is sent as ordinary Ethernet frames using the **Precision Time Protocol (PTP, IEEE 1588)**. That part of the fronthaul is called the **S-plane**, the synchronization plane.

The problem. When timing goes wrong, an operator has to choose between two opposite actions. If it is an honest fault, such as a grandmaster losing GNSS, the right move is to *ride it out in holdover*. If it is an attack, such as a rogue clock lying about time, the right move is to *isolate the source immediately*. The symptom ("the clock is wrong") looks the same in both cases. A detector that only says "anomaly" is not enough.

What we built. A software testbed of **six real linuxptp 4.0 daemons** in Linux network namespaces, using the **ITU-T G.8275.1 telecom profile**. Two grandmasters and a boundary clock feed three radio units. We ran **14 scenarios × 12 randomised replicates = 168 live runs**, producing **1,451,909 packet records** with **56 fields each**.

How we decide. Each run is judged by a **decision rule**. It checks what the standards say is legal, plus what the operator has provisioned: which clocks are allowed, which boundary clock is expected, and whether a maintenance window is open. It returns **ATTACK** (isolate), **BENIGN** (tolerate or holdover) or **UNKNOWN** (escalate to a human). The rule was fingerprinted with SHA-256 before the runs so it could not be tuned to the answers.

0.990

attack sensitivity, revised rule v3 (pre-frozen rule: 0.625)

0.783

benign specificity (0.979 without the one known failure)

12 / 12

correct UNKNOWN on the deliberately ambiguous case

84 / 96

attack runs with the exact fault named (0.875)

Rule versus machine learning. On 56 held-out runs the rule scored **51/56**. A Random Forest scored **24/56**. A "predictor" that always answers BENIGN scored **20/56**. The ML model is not statistically different from that trivial baseline (McNemar $p = 0.219$). The reason is observability: it only had 6 of its 28 planned features, because the testbed has no GNSS, SyncE or oscillator data.

What is not done. No fault prediction, no digital-twin validation, and no automated healing in the 168-run campaign. A single port-failover action was executed in an earlier 13 September pilot (5/5 against 0/5 controls). The one open benign failure, a planned boundary-clock replacement read as an attack, has a known, located cause.

The one sentence to remember: *we built a controlled, standards-grounded benchmark that tells timing attacks apart from honest timing faults. On today's telemetry a transparent rule does this far better than a learned model, so the model should not be given control of recovery yet.*

PART I

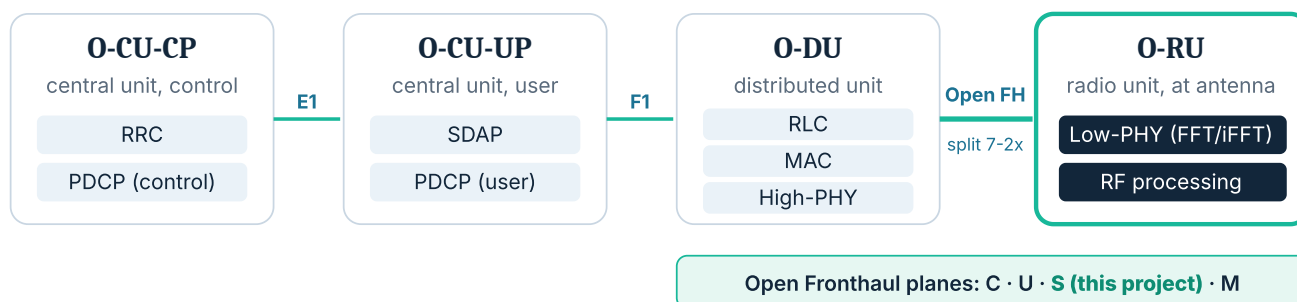
The story of the project

From what a base station is, to how time moves across a cable, to the testbed, the scenarios, the rule and the model. Read this slowly once and the deck will make sense on its own.

A base station broken into pieces

Picture an older mobile base station as one sealed cabinet from one vendor. Everything happened inside it: the radio, the signal processing and the protocol stack. **O-RAN** opened that cabinet. It split the base station into separate units with standard interfaces between them, so that different vendors can build different parts.

There are four units you need to know. The names and what each one hosts come from ETSI TS 103 982, the O-RAN Architecture Description, clause 3.1.



The O-RAN functional split (ETSI TS 103 982 V8.0.0, clauses 3.1, 6.3.3–6.3.6, 6.4.7). The project works on the S-plane of the Open Fronthaul, between the O-DU side and the O-RU.

The link between the O-DU and the O-RU is the **Open Fronthaul**. It uses split option **7-2x**, which means the physical layer itself is cut in two: High-PHY stays in the O-DU and Low-PHY moves into the O-RU. Four kinds of traffic share this link, called **planes**:

- **C-plane** (control): tells the radio what to do with the next slots.
- **U-plane** (user): the actual radio samples, meaning your data.
- **S-plane** (synchronization): **time and frequency**. This is where our project lives.
- **M-plane** (management): configuring and monitoring the radio unit.

A detail reviewers like to test. The E2 interface, which the RIC uses to collect KPIs, ends at the O-CU-CP, O-CU-UP and O-DU. It does *not* reach the O-RU (clauses 6.3.3–6.3.6). So the usual "RAN KPIs over E2" are not available at the S-plane. Our evidence has to come from the timing protocol itself.

How the O-RU gets its time: LLS-C1 to C4

O-RAN defines four ways to deliver time to the radio unit. We quote them from Armstrong's ATIS 2023 tutorial, which cites O-RAN.WG4.CUS.0 v06.00. The deck says this is a secondary source.

CONFIG	WHO GIVES THE O-RU ITS TIME	WHERE AN ATTACKER CAN ACT
LLS-C1	The O-DU, over a direct link to the O-RU.	A compromised O-DU or the O-DU–O-RU link.
LLS-C2	The O-DU, through one or more Ethernet switches.	As C1, plus any device on the switched fronthaul.
LLS-C3	A grandmaster (PRTC/T-GM) in the network. The O-DU is <i>not</i> in the timing chain.	Grandmaster, boundary clocks and switches on the path.
LLS-C4	A local GNSS receiver at the O-RU. No network timing at all.	No PTP exposure. GNSS jamming or spoofing instead.

Our testbed has the **LLS-C3 shape**: grandmaster → boundary clock → radio units, with no O-DU in the timing chain. TIMESAFE's real production test also used LLS-C3. The O-DU itself is not emulated.

Why time matters so much

Imagine several radio units sharing one area. If they transmit at slightly different instants, their signals collide instead of combining. Techniques like MIMO and carrier aggregation only work if the radios behave as one instrument, so they must share the same idea of "now".

130 ns

maximum relative error between radio units in a cluster (O-RAN timing category A). Split equally, that is about ± 65 ns per O-RU against a common reference.

$\pm 1.5 \mu\text{s}$

absolute time-error budget at reference point E, the end application, against a common time standard (ITU-T G.8271.1). The network's share at point C is 1,100 ns.

$\approx 2 \text{ s}$

observed time from attack start to O-RU crash in one published experiment (TIMESAFE, ACM TOPS 2025). This is a measurement, not a standard.

Behind these sit the 3GPP air-interface limits: cell phase synchronization better than **3 μs** (TS 38.133), and time-alignment error of **65 ns** for MIMO and **260 ns** for intra-band contiguous carrier aggregation (TS 38.104, clause 9.6.3.2).

Now the uncomfortable part. Time reaches the radio as **ordinary Ethernet frames**. Any device that can put frames on that wire can:

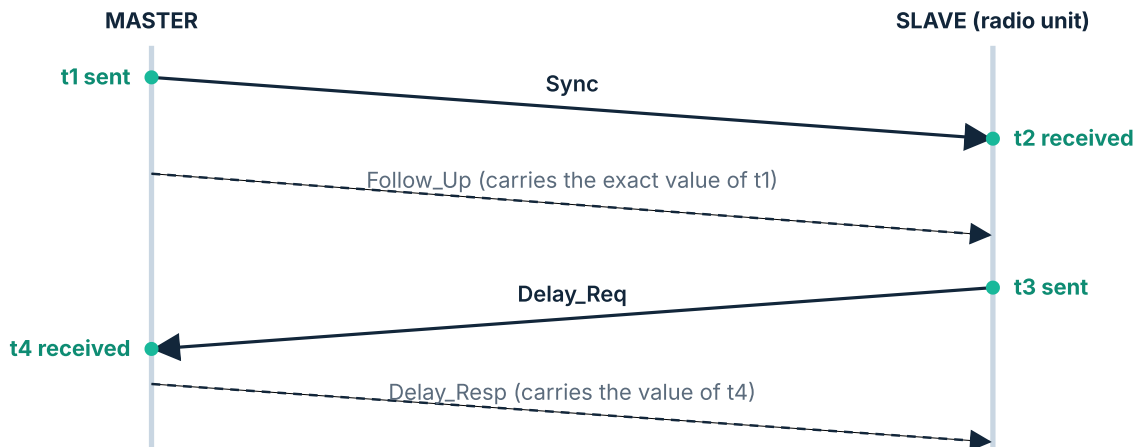
- pretend to be a better time source,
- forge the time values inside the messages, or
- quietly drop the real messages.

So the S-plane is two things at once: a **performance boundary** (stay within nanoseconds) and an **attack surface**. TIMESAFE showed what that means in practice. In their production-style network, with switch PTP ports in dynamic role, the radio unit crashed about two seconds after a spoofing attack began. The 5G cell dropped, and the radio needed a manual reboot.

How time travels: PTP, explained slowly

PTP, the **Precision Time Protocol**, is defined in **IEEE 1588-2019**. Its job is simple to state. One clock, the **master**, knows the time. Another clock, the **slave**, wants to match it. The difficulty is that every message takes time to travel, so the slave must work out how far behind it is *and* how long the cable takes.

The four timestamps



The two-step exchange. Sync and Follow_Up carry time from master to slave. Delay_Req and Delay_Resp measure the path in the other direction.

After one exchange the slave knows four numbers. Assuming the cable takes the same time in both directions:

$$\text{mean path delay} = [(t2 - t1) + (t4 - t3)] / 2$$

$$\text{offset from master} = [(t2 - t1) - (t4 - t3)] / 2$$

The slave's **servo** then nudges its own clock to drive that offset toward zero. In the linuxptp program **ptp4l**, every correction shows up as a log line. Here is a real one from our baseline run 1, radio unit 1:

```
ptp4l[517.414]: master offset 1191 s0 freq -1547 path delay 5313
```

Read it as: offset 1,191 ns; servo state **s0** (not steering); frequency adjustment -1,547 ppb; path delay 5,313 ns. In our testbed the servo state stays **s0** in every radio-unit log, because every daemon runs with `free_running 1`. Chapter 7 explains why.

Who is the master? The election

Several clocks may offer to be the master. Each one broadcasts an **Announce** message describing itself, and every clock runs the same election, the **Best Master Clock Algorithm (BMCA)**, to pick the best. The Announce fields that matter for this project:

FIELD	MEANING, SIMPLY	WHY IT MATTERS HERE
clockIdentity	The clock's 8-byte name.	Checked against the operator's allow-list.
priority1 / priority2	Administrator preferences.	G.8275.1 fixes priority1 at 128. Any other value is illegal.
clockClass	How good the source is. 6 = locked to a primary reference. 7 = in holdover within spec. 248 = default.	A drop from 6 to 7 is the honest signature of GNSS loss.
stepsRemoved	How many clocks sit between this one and the grandmaster.	An inserted rogue relay adds a step.
sequenceId	A counter that must keep increasing.	Replayed old frames make it go backwards.

**timeProperties
flags**

UTC offset (37 s), leap-second flags, traceability.

Faking these shifts time by a whole second.

The chain of clocks

The top clock is the **grandmaster** (GM), usually locked to GNSS. A **boundary clock** (BC) is a middle clock: a slave on its upstream port and a master on its downstream port, re-serving time to the next segment. At the bottom are the **radio units**, which are slaves.

The telecom rules: ITU-T G.8275.1

IEEE 1588 is a general toolkit. A **profile** fixes the settings for one industry. Telecom uses **ITU-T G.8275.1** (full timing support from the network). Our testbed ran it exactly as configured and read back from the running system:

domain 24

priority1 128

Sync 16 / s

Announce 8 / s

alternate BMCA (G.8275.x)

L2 multicast 01-1B-19-00-00-00

software timestamping

Why this matters for security. Because the profile fixes these values, a lot of "is this legal?" becomes a yes/no check against the standard rather than a guessed threshold. That is the backbone of our decision rule.

The question at the heart of the project

Here is the scene the whole project is built around. A radio unit's time source suddenly changes, or its time starts to look wrong. Two very different stories could explain it.

BENIGN FAULT

Example: the grandmaster loses its GNSS signal and honestly says so, moving clockClass from 6 to 7.

Right response: ride it out in **holdover**.

Isolating the node here would *create* an outage that was not going to happen.

ATTACK

Example: a rogue clock sends forged Announce messages and wins the election.

Right response: **isolate the source** immediately.

Waiting in holdover lets the attacker keep steering time.

Same symptom · opposite correct response · the symptom alone cannot decide.

So we needed three possible answers, not two:

VERDICT	WHEN	RECOMMENDED RESPONSE
ATTACK	Evidence is illegal under the standard, or unauthorised under the operator's provisioning.	ISOLATE the source
BENIGN	Only provisioned clocks are involved, nothing is illegal, and the declared state agrees with what is observed.	TOLERATE / HOLDOVER
UNKNOWN	The packets genuinely cannot reveal intent.	ESCALATE to an operator

The key idea, in one line: **for a benign fault, the declaration and the observation agree; for an attack, they disagree.** When a grandmaster changes, the deciding test is not the timing numbers. It is two questions: *is the new clock on the operator's allow-list, and is a maintenance window open?*

Why the S-plane was the right piece of the worklet to study

1. **Highest consequence.** A timing failure can become a service failure within seconds.
2. **Genuine ambiguity.** Some benign faults and attacks produce overlapping symptoms.
3. **Constrained by standards.** IEEE 1588-2019 and G.8275.1 give objective legality criteria.
4. **Reproducible in software.** Real PTP daemons can be attacked on a virtual wire without a proprietary RAN stack.

Who attacks, and how they get in

The reviewer asked "who does these attacks?" The honest answer is that the standards do not name people. They describe attackers by **capability and position**. We used two sources.

THREAT AGENTS · ETSI TR 104 106, cl. 7.2

Insiders · nation-state actors · cyber-criminals · hackers ·
cyber-terrorists · script kiddies.

ATTACKER CLASSES · IETF RFC 7384, cl. 3.1

Internal: on a trusted segment, or holds the keys. **External:** no keys. **Man-in-the-middle:** can intercept and modify. **Injector:** not in the path, only sends packets.

How they reach the fronthaul (TIMESAFE section 4; ETSI T-FRHAUL-02):

- remote access widened by multi-vendor, disaggregated equipment;
- physical access to edge cell sites;
- supply-chain backdoors in DU or RU equipment;
- unauthorised access to the Open Fronthaul Ethernet interface.

TIMESAFE puts it plainly: once an attacker controls a device on the switched network, they can observe, replay or inject PTP packets.

Mapping our attacks. Six of our eight attacks fall in RFC 7384 rows that need an **internal** attacker: rogue master (A1, A8), spoofing (A2, C3), replay (A3) and time-protocol DoS (A5). Interception (C1) can also be done by an **external** man-in-the-middle. Malformed input (C2) falls under "exploiting vulnerabilities" (cl. 3.2.11), which the RFC table does not classify.

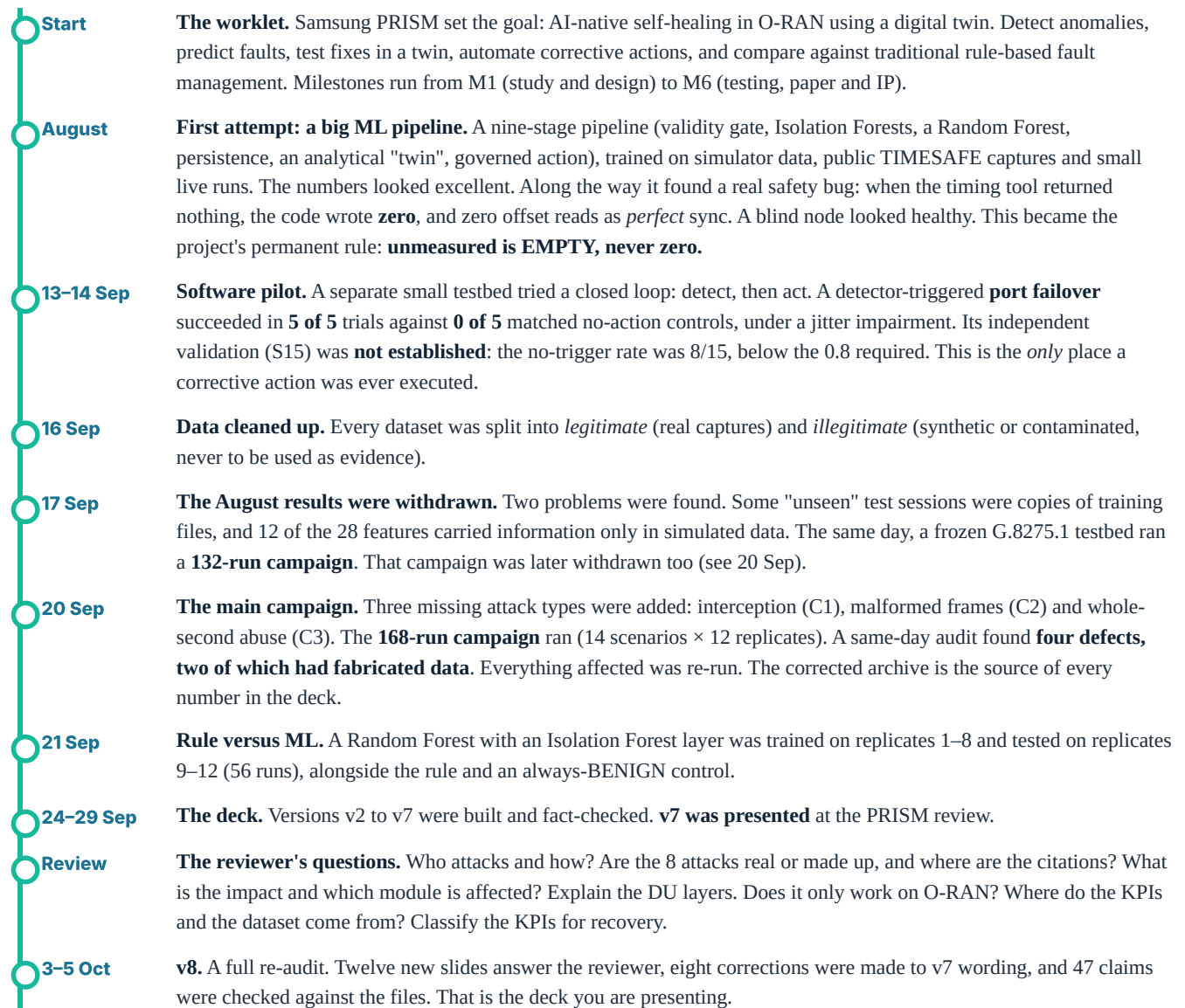
Were the attacks invented?

No. Seven of the eight follow a published threat or test. C3 is our own instance of the RFC 7384 spoofing class: we did not find a published test of that exact field combination.

ID	ATTACK	PUBLISHED THREAT	PUBLISHED TEST OR STUDY
A1	Rogue grandmaster	RFC 7384 §3.2.4; T-SPLANE-03	TS 104 105 §11.1.5.2.2; TIMESAFE spoofing
A2	Sync / Follow_Up spoofing	RFC 7384 §3.2.2	No ETSI test of this exact form
A3	Replay	RFC 7384 §3.2.3	TIMESAFE replay attack
A5	DoS / PTP flooding	RFC 7384 §3.2.9; T-SPLANE-01	TS 104 105 §11.1.5.1.1, §24.2.1.1
A8	Rogue boundary clock	RFC 7384 §3.2.4 (in MITM position)	Adapted: no test places the rogue as a relay
C1	Interception and removal	RFC 7384 §3.2.5; T-SPLANE-04	TS 104 105 §11.1.5.3.1
C2	Malformed frames	RFC 7384 §3.2.11	TS 104 105 §24.2.1.2
C3	Whole-second field abuse	RFC 7384 §3.2.2 (false time)	None found; fields per IEEE 1588-2019 §8.2.4

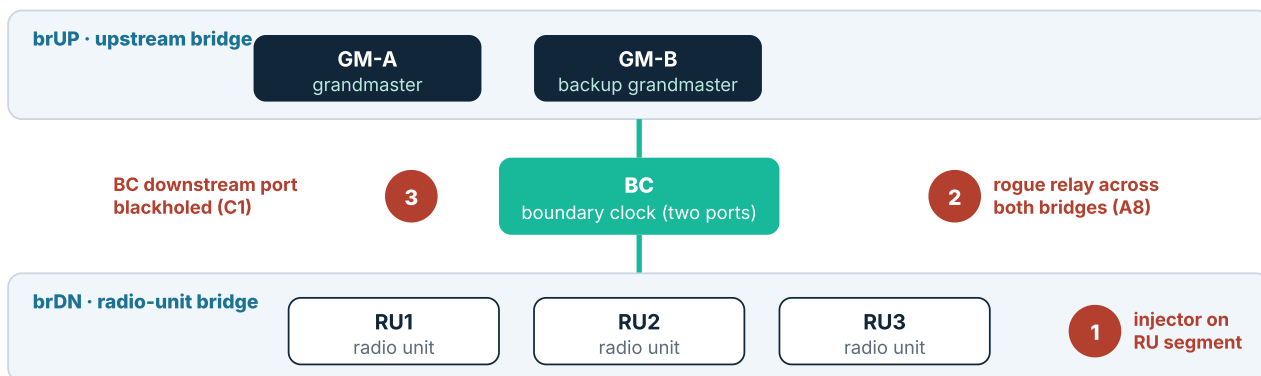
The journey: what we did, in order

The project did not go in a straight line. Knowing the order explains why the final design looks the way it does: every safeguard in it was added because something earlier went wrong.



The testbed we built

Everything runs on one Linux host. Each "machine" is a **network namespace**, an isolated network stack with its own interfaces. They are wired together with virtual cables (veth pairs) into two software bridges. Inside each namespace runs a **genuine linuxptp 4.0 ptp4l daemon**. The election, the message cadence and the servo are all executed by real protocol software, not simulated.



Six real ptp4l daemons on two bridges. Red markers show where attacks enter. Position 1 covers A1, A2, A3, A5, C2 and C3; position 2 is A8; position 3 is C1. All read from the campaign scripts.

What was recorded in every run

EVIDENCE	WHAT IT IS	USED BY
Packet capture	tcpdump on both bridges, then a deep extractor writes 56 fields per PTP frame (header, flags, correctionField, identities, sequenceId, Announce data set, G.8275.1 conformance flags). The raw PCAPs (about 222 MB) were not kept in the packaged archive; the extracted CSVs were.	The rule
Servo logs	ptp4l output from all six daemons: offset, frequency, path delay, port-state changes.	The ML model
pmc queries	Management queries for 7 data sets per node, every second. They returned no data in any run (40,320 unanswered requests). The likely cause is pmc called without <code>-d 24</code> while ptp4l used domain 24; this was not re-run.	Nobody
context.json	The ground-truth label plus the operator's provisioning: allow-list, expected boundary clock, maintenance-window state.	The rule (as context)
Three verdicts	The base rule, v2 and v3, each scored on the same capture.	Evaluation

The honest limits of the platform

- **No PTP hardware clock** (`/dev/ptp*` is empty), so timestamps are taken in software. Their noise is at the microsecond scale, while the targets are nanoseconds. **No claim is made about absolute time-error size.**
- **One shared oscillator.** All six namespaces use the same host clock, and every daemon runs with `free_running 1`. The offset is *measured* but the clock is *never steered*, and the host clock is never touched. So the "false time" impact of an attack is seen at the protocol level, not as a real clock being pulled off.
- **No loadable kernel modules** (no netem). Congestion was built with a `tbfb` bottleneck instead.
- **No GNSS receiver and no SyncE PHY.** GNSS spoofing and jamming, holdover, SyncE and oscillator-drift classes cannot be produced here.

The fourteen scenarios

Each scenario was run 12 times with randomised parameters (identity, priority, burst size, gaps, onset). The random draws are seeded from the replicate number, so every run can be reproduced exactly. The important design idea is that **attacks and their benign look-alikes go through the same packet path**: the evidence separating them is context, not mechanics.

Eight attacks

ID	WHAT WE DID, PLAINLY	WHAT GIVES IT AWAY
A1 Rogue grandmaster	Started a second real ptp4l with an identity <i>not</i> on the allow-list and a better data set (random priority2 and clockClass). It wins the election fairly.	Its identity is not provisioned and no maintenance window is open.
A2 Sync spoofing	Hand-built Sync/Follow_Up frames (with Scapy) from an unknown source, in bursts, with no Announce.	Unknown sender pushing timing content.
A3 Replay	Captured real frames mid-run with tcpdump and sent them again with tcpreplay.	sequenceId goes backwards; timestamps are stale.
A5 Flooding	Bursts of 300 PTP frames from an unauthorised source.	Rate far above the profile; about 38,000 records per run against about 6,000 at baseline.
A8 Rogue boundary clock	A two-port ptp4l that slaves to the real grandmaster and re-advertises downstream with stepsRemoved +1. It relays genuine time.	An unauthorised clock inserted in the timing path.
C1 Interception	Shut the boundary clock's downstream port for 60–75 % of the injection window. This removes <i>every</i> frame, so it is a coarse version of "selective" removal.	Delivered rate falls below half the rate the source declares.
C2 Malformed frames	Frames with versionPTP 3, a reserved message type and a too-short length.	Structurally illegal under IEEE 1588-2019.
C3 Whole-second abuse	Announce frames with leap61 set, UTC offset faked to 0 (true value 37) and traceability flags stripped.	Time properties are internally inconsistent.

Five benign scenarios, and one deliberately ambiguous case

ID	WHAT HAPPENED	WHY IT IS HONEST
baseline	Nothing injected. GM-A serves all three radio units through the boundary clock.	Healthy control.
B2 Planned GM changeover	GM-A is withdrawn during an open maintenance window; allow-listed GM-B takes over.	In packets it looks <i>exactly</i> like A1. Only the allow-list and the maintenance state differ.
B3 Congestion	A 1 Mbit tbf bottleneck with competing non-PTP traffic. Sync delay variation rises from 0.07 ms to 5.6–857 ms, with no loss.	Delay grows but no field becomes illegal.
B7 Topology change	A radio-unit link is bounced and re-homed during maintenance.	stepsRemoved and path delay change consistently.
B_bc_replacement	A provisioned replacement boundary clock is swapped in during maintenance.	Structurally identical to A8, but planned and provisioned.
B_unplanned_failover (ambiguous)	GM-A fails with <i>no</i> maintenance window; allow-listed GM-B is elected.	Packets cannot tell a real failure from an attack that knocked out GM-A. The correct answer is UNKNOWN .

These 14 come from a 16-class fault catalogue. Eight classes were executed in software. Three have software detection logic but would need hardware to measure their physical impact (A4 delay attack, A8, B5 asymmetry). Five need hardware and could not be produced (A6 GNSS spoofing, A7 GNSS jamming, B1 holdover, B4 SyncE, B6 oscillator drift). C1–C3 are campaign identifiers taken from O-RAN WG11 tests and sit alongside the catalogue.

From a packet to a verdict

01 · WIRE

real PTP frames
from 6 daemons

02 · CAPTURE

tcpdump on
brUP and brDN

03 · EXTRACT

56 fields per frame
unseen = EMPTY

04 · RULE

legality + operator
context, in order

05 · VERDICT

ATTACK · BENIGN
UNKNOWN + clause

A run produces a few thousand PTP frames. The extractor turns each frame into one row of 56 columns. Then the rule reads all the rows for that run and gives one verdict, **recording which clause fired**, so every decision can be traced back to its evidence.

A real example: A1, replicate 1

From the traceability workbook. The operator's allow-list is `...00000a` and `...00000b` (GM-A and GM-B), and no maintenance window is open.

ADVERTISED GRANDMASTER	ANNOUNCE PACKETS	ON ALLOW-LIST?
020000fffe00000a (GM-A)	371	yes
020000fffe000042 (rogue)	157	NO
020000fffe000001 (BC, start-up)	2	no

Look at what the rogue sent: priority1 = **128** (legal), priority2 = 1, clockClass = **6**. The legitimate grandmaster advertised clockClass 248. **Every field in the rogue's frames is legal**, and clockClass 6 is *better* than 248, so it wins the election fairly, without breaking any rule of the standard. The only thing that gives it away is that its identity is not provisioned and no maintenance is open.

This is the project's core argument in raw data. Standards alone cannot catch the most dangerous attack. Standards plus the operator's provisioned intent can. The clause that fired: *"Unauthorised clock advertising ITSELF as grandmaster (not in the provisioned allow-list)"*.

The decision rule, and why it was frozen

The rule is a list of ordered checks. The base version (`decision_rule.py`) goes, in order:

1. Message rate flood → **A5**
2. sequenceId goes backwards → **A3**
3. Unauthorised clock relaying in the path → **A8**
4. Unauthorised clock announcing itself as grandmaster → **A1**
5. Illegal clockClass, priority1 $\neq$ 128, stepsRemoved $\geq$ 255, alternateMaster flag → **ATTACK**
6. Grandmaster not on allow-list → **A1**
7. Unknown sender of timing messages → **A2**
8. Out-of-profile domain, multicast or cadence → **A5**
9. Grandmaster changed *with* maintenance open → **BENIGN** (B2)
10. Grandmaster changed *without* maintenance → **UNKNOWN**

Version 3 adds three detectors **on top**, never removing anything:

- **D1, rate starvation (C1)**: the delivered message rate is below half the rate the source itself declares (ratio below 0.500). It has a minimum-frame floor and a persistence requirement.
- **D2, malformed frames (C2)**: versionPTP, reserved message types, impossible lengths.
- **D3, whole-second abuse (C3)**: leap flags, UTC offset and traceability that do not agree.

Why "frozen" matters

If you write a rule after looking at the answers, it will of course get the answers right. To stop that, the base rule was **SHA-256 hashed before any run started**. The hash is a fingerprint: change one character and it changes, so anyone can check the rule was not edited afterwards. Expected verdicts for each scenario were also written down in advance.

RULE	HASH (START)	STATUS
Base rule	c362e110...	Frozen before the campaign. This is the honest pre-registered result.
v2	e32eb65b...	Frozen at 11:41:31Z on 20 Sep, before the campaign began at 11:43:15Z. Gives verdicts identical to v3 on all 168 runs.
v3 (primary)	aa9417b7...	A disclosed revision after observation. It fixed v2 false positives on upstream captures and is "additive-only".

One nuance to be ready for. v2 was frozen before the 168 runs, but it was developed after earlier same-day captures of the same C-scenario designs. That is why the deck always shows both figures: **0.625** for the pre-frozen base rule and **0.990** for the revised rule, on the same captures.

Additive-only, checked per run: there are zero runs where the base rule said **ATTACK** and v3 did not. There are 35 escalations, all on C1, C2 and C3.

The machine-learning arm

The worklet asks for a comparison between the AI approach and traditional rule-based fault management. So we put a learned model on the same runs.

ARM A · Rule

56 packet fields plus operator context. No training. Frozen.

ARM B · ML

Random Forest (90 trees, depth 6, balanced classes, seed 1588) with an Isolation Forest novelty layer for abstention. Inputs: ptp4l servo telemetry only.

Control

Always answers BENIGN. If the model cannot beat this, it has learned nothing useful.

The split. Trained on replicates 1–8 and tested on replicates 9–12, which gives 56 held-out runs. The split is by whole run, never by window, so no piece of a test run is ever seen in training. Hyperparameters and the split were fixed before scoring.

Why the model was starved. It was designed for 28 features in six groups (timing, protocol regularity, message rate, BMCA, time source, oscillator). In this testbed only **6 of 28** carried data: offset_mean, offset_std, offset_abs_max, path_delay_mean, pdv_std and holdover_rate. Holdover rate had zero importance, so **five features carried all the signal**, all of them timing magnitudes. GNSS, SyncE, oscillator and every protocol-legality feature simply do not exist in ptp4l servo logs.

A worse twist. Some attacks starve the very telemetry the model needs. Servo samples average **88.0** per baseline run, **44.5** under interception and **28.0** under whole-second abuse. An attack that suppresses timing messages also suppresses the evidence a telemetry-based detector relies on. The rule is unaffected, because it reads packet legality and rate deficit directly.

Being honest with ourselves: the audit

On 20 September, after the campaign ran, a cross-check audit of our own apparatus found four defects. Two of them had produced fake data. This chapter is worth knowing well, because it is the strongest proof that the remaining numbers can be trusted.

DEFECT	WHAT WENT WRONG	FIX
36 "replicates" were copies	A runner script lacked the execute permission. The error (exit code 126) was hidden by output redirection, and a stale capture then passed the packet-count check. Three captures (C1, C2, C3) were each copied twelve times. Proof: one SHA-256 across twelve "replicates", and runs finishing in under 1 s instead of about 48 s.	Fixed and re-run
B3 congestion never injected	The kernel has no netem, and the failure was swallowed by a shell fallback. "Congestion" was really a baseline with a congestion label.	Rebuilt with tbf plus competing traffic
v2 false positives	The rate check had no guard for start-up. A boundary clock sending three frames during election read as "rate-starved", which turned a benign failover into ATTACK and destroyed a correct UNKNOWN.	v3 adds a minimum-frame floor and persistence
Evaluator overstated	Pooled metrics moved with run count, additive-only was checked on two summary numbers, and abstention was credited on error paths.	Replaced by a per-scenario evaluator (v4) with per-run checks

Because the B3 defect also affected the earlier 132-run campaign of 17 September, that campaign was withdrawn in full. **Every figure in the deck comes from the corrected 168-run archive** (`splane_campaign_CORRECTED_2026-09-20.tgz`, 1,096 files, sha256 6149b4fb...).

How to say it: "We audited our own pipeline, found two defects that had fabricated data, re-ran everything affected, and report only the corrected campaign. Disclosing that is part of the result."

PART II

The deck, slide by slide

All 42 slides of ORAN_SPlane_PRISM_Review_v8. For each: what it shows, a line you can say, and what to keep ready if asked. Slides marked NEW were added to answer the reviewer.

How the 42 slides are organised

The deck follows the order a listener needs: what the system is, why timing is fragile, who attacks, how we tested, what we found, and what is left. The twelve new slides are placed exactly where each reviewer question arises, and slide 3 maps every question to its answer.

SLIDES	BLOCK	PURPOSE
1–3	Opening and review map	Worklet, title, and where each reviewer question is answered
4–8	Architecture and problem	O-RAN split, LLS-C1...C4, timing requirements, the attack–fault ambiguity
9–12	Threat model	Who attacks, entry points, provenance of each attack, impact
13–16	Background and standards	Literature, gaps, positioning, standards used
17–23	Testbed, pipeline, KPIs, dataset	What was built, what was measured, where data came from
24–31	Scenarios and method	Fault injection, the 14 scenarios, decision logic, freezing, the audit
32–35	Results	Per-scenario, aggregate, rule versus ML, why ML underperformed
36–39	Limits and status	Detection boundary, limitations, worklet status, remaining work
40–42	Close	Applicability beyond O-RAN, summary, references

Time management. 42 slides will not fit an 8–10 minute slot at full depth. A workable route: spend about a minute on 4–8, two minutes on 9–12 (the reviewer's main concern), one minute on 17–22, two minutes on 32–35, and close on 37–38 and 41. Keep the rest as backup to jump to when asked.

AI-Native Self-Healing Open-Radio Access Network (O-RAN) Network using Digital Twin

In native self-healing O-RAN Network using Digital Twin proposed an autonomous network management framework that combines AI-based anomaly detection with a digital twin to predict, validate, and execute RAN faults in a closed loop. The project covers network anomaly, faster recovery, and lower operational effort for future 5G networks.

Problem Statement

5G networks are becoming more complex and distributed, which makes manual fault diagnosis and recovery slower and less reliable. Traditional static alarms and rule-based responses often react too late, require human intervention, and may not identify the true root cause of a problem. The project aims to solve this by combining AI and a digital twin to create a closed-loop self-healing mechanism that can sense, predict, decide, and verify recovery actions.

Key Objectives

- Detect anomalies in O-RAN KPIs using AI-based monitoring.
- Predict faults before they cause service disruption.
- Automate corrective actions such as parameter tuning, reconfiguring, or resource allocation.
- Compare the AI-native approach with traditional fault management methods.

References

1. Open RAN Architecture: Network Splitting (Open RAN, 2020).
2. Digital Twin in 5G Networks: A Review (IEEE Communications Surveys & Tutorials, 2021).
3. AI-based Anomaly Detection in 5G Networks (IEEE Transactions on Network Intelligence, 2022).
4. Self-Healing Networks: A Review (IEEE Communications Surveys & Tutorials, 2021).
5. Digital Twin for 5G Networks: A Review (IEEE Communications Surveys & Tutorials, 2021).

Overview

Build a closed-loop system where a digital twin mirrors the live O-RAN environment, detects anomalies, predicts faults, and recommends or triggers healing actions before service impact grows. Research on O-RAN digital twins and AI-driven self-healing highlights this as a practical path toward autonomous RAN operations.

Expected Outcome

- A conceptual architecture for AI-native self-healing in O-RAN.
- A digital twin-based testbed for fault injection and recovery validation.
- AI models for anomaly detection or predictive fault identification.
- Faster fault recovery and reduced downtime compared with manual or rule-based methods.
- Evaluation results showing improvements in availability, recovery time, and operational efficiency.

Key Milestones

- M1:** KPI and fault taxonomy. Identify KPIs, anomaly types, and healing targets.
- M2:** Digital twin setup. Build or simulate the O-RAN env and fault injection environment.
- M3:** AI model development. Implement anomaly detection or predictive fault models.
- M4:** Training and evaluation. Measure detection accuracy, recovery time, and service continuity.
- M5:** Project & IP Draft. (Publication + Patent disclosure) Final documentation. Prepare results, findings, and future enhancement ideas.

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1 OPENING

Worklet definition (PRISM)

SHOWS The original Samsung PRISM worklet, reproduced unchanged: the problem statement, objectives, expected outcomes and milestones M1–M6.

SAY This is the worklet we were given. Our work is one sub-scope of it: timing security on the Open Fronthaul S-plane.

IF ASKED Objective 5 (compare with rule-based) is answered on slide 34. Objectives 2 (prediction) and 4 (automated correction) are stated as not yet done on slide 38.

Open Fronthaul S-Plane
Timing Security

Sub-scope of the AI-Native Self-Healing O-RAN Network using a Digital Twin worklet

TITLE

Tarensya Kumar - Raghu Ram K - Manipal Jaseenth Kumar

MENTOR

Mr. Bikas Singh (NovaThink Tech)
Dr. Navin Kumar (Amrita Vishwa Vidyapeetham)

AMRITA SCHOOL OF ENGINEERING, BANGALORE - SAMSUNG PRISM

168 live runs - 14 fault scenarios - 12 randomised replicates

2 OPENING

Title

SHOWS Project title, team, mentors, and the campaign in one line: 168 live runs, 14 scenarios, 12 randomised replicates.

SAY We report what was implemented, what data was generated, what was inferred, and what remains.

REVIEW RESPONSE

Reviewer Questions and Where Each Is Answered

Questions borrowed from the reviewer's handwritten notes on the PRISM review presentation | Every row slide carries numbered references [x] listed at the end of the deck

REVIEWER QUESTION	ANSWERED ON
What kind of attack, by whom, and how? Who does these attacks? Source of attack?	Threat model and entry points: slides 9, 10
Were the 8 attacks existing or made up? Where were they referred from? Citations?	Attack provenance: slide 11; references: slide 42
What are the impacts? Which module will these attacks affect?	Impact and affected component: slide 12
Explain the DU layer and the architecture	Functional split and timing configurations: slides 5, 6
Does it only work on O-RAN, or on other architectures?	Applicability: slide 40
From which model did we get the KPIs?	KPI sources: slide 20
Where did we get the dataset?	Dataset provenance: slide 23
Classify the KPIs so that we can proceed with recovery	KPI classification for recovery: slide 21

ALSO CHANGED Presentation: answers are placed where each question arises. Corrections to v7 are listed in the speaker notes of this slide.

3 REVIEW RESPONSE

Reviewer questions and where each is answered

SHOWS Every handwritten reviewer question, mapped to the slide that answers it. The speaker notes list the eight corrections made to v7.

SAY After the last review we restructured the deck so each of your questions is answered where it naturally arises. This slide is the map.

IF ASKED Corrections to v7: raw PCAPs not archived; pmc returned no data; free_running 1; A1 wording; $\pm 1.5 \mu\text{s}$ at point E; B_bc_replacement cause; corrective action only in the pilot; v2 frozen before the campaign.

SYSTEM ARCHITECTURE AND PROJECT BOUNDARY

End-to-End O-RAN Architecture and S-Plane Project Boundary

O-RAN Alliance architecture description (ETSI TS 103 982 V8.0.0 | O-RAN WG2 C-SCG, User and Synchronization Plane Specification | 3GPP TS 38.401 NG-RAN architecture)

PROJECT FOCUS: OPEN FRONTHAUL S-PLANE
PTP timing end-to-end observed on O-RU and O-DU. SyncE was outside this software testbed.

Validation from live packet path operations, reduced timing end-to-end and distribution, narrowing scope: digital twin architecture, fault prediction, automated recovery and outcome measurement.

4 ARCHITECTURE

End-to-end O-RAN architecture and project boundary

SHOWS The full O-RAN picture (SMO, RICs, O-CU, O-DU, O-RU, core) with the timing chain beneath it and the S-plane region highlighted as our scope.

SAY Our work sits exactly here: the Open Fronthaul S-plane, where timing reaches the radio unit.

IF ASKED E2 does not reach the O-RU, so RIC KPIs do not exist at our boundary. SyncE was outside this software testbed.

SYSTEM ARCHITECTURE - PROTOCOL LAYERS

O-RAN Functional Split: What the O-DU and O-RU Host

ETSI TS 103 982 V8.0.0 (2024-01), O-RAN Architecture Description, clauses 3.1, 6.3.3–6.3.6, 6.4.7 [5] | 3GPP TS 38.401 NG-RAN architecture [11]

OPEN FRONTHAUL PLANES (TS 103 982 V8.0.0, 6.4.7)

O-Plane: Control | U-Plane: User | S-Plane: Synchronization - This project | M-Plane: Management

WHY THE O-RU?
Radio requirements are measured at the base-station antenna [ETSI]. In O-RAN the antenna side is the O-RU (Low PHY + RF). The S-plane is how the O-RU receives time, so it is where this project works.

5 ARCHITECTURE NEW

O-RAN functional split: what the O-DU and O-RU host

SHOWS Layer by layer: O-CU-CP (RRC, PDCP-C), O-CU-UP (SDAP, PDCP-U), O-DU (RLC, MAC, High-PHY), O-RU (Low-PHY, RF), and the four Open Fronthaul planes.

SAY Radio requirements are measured at the antenna, and in O-RAN the antenna side is the O-RU. The S-plane is how the O-RU receives time, so that is where we work.

IF ASKED Source: ETSI TS 103 982 V8.0.0, clauses 3.1, 6.3.3–6.3.6 and 6.4.7. Split option 7-2x.

How the O-RU receives time: LLS-C1 to LLS-C4

SAY Our testbed has the LLS-C3 structure: grandmaster to boundary clock to radio units, with no O-DU in the timing chain. TIMESAFE's production test was also LLS-C3.

7 PROBLEM STUDY

Timing requirements and security exposure

SAY Time travels as ordinary Ethernet frames, so anything that can put frames on that path can lie about time. The S-plane is both a performance limit and an attack surface.

8 PROBLEM STUDY

Attack-fault ambiguity and sub-scope selection

SHOWS A benign GNSS loss and a rogue master look the same ('the clock is wrong') but need opposite responses: holdover versus isolate. Four reasons for choosing the S-plane.

IF ASKED Why not the whole RAN? Scoping to the S-plane is what made rigorous, repeatable measurement possible in the time available.

9 THREAT MODEL NEW

Who attacks the S-plane: agents and classes

SHOWS Threat agents from ETSI TR 104 106 cl. 7.2; attacker classes from RFC 7384 cl. 3.1; access paths from TIMESAFE and T-FRHAUL-02.

IF ASKED We do not attribute any attack to a named real-world actor, and no public source used here does.

10 THREAT MODEL NEW

Attack entry points on the testbed timing path

SHOWS Three attacker positions on the testbed diagram and exactly how each attack was launched (second ptp4l, Scapy frames, tcpdump plus tcpreplay, port down).

SAY Position 1 is an injector on the radio segment; position 2 is a rogue relay across both bridges; position 3 is in-path at the boundary-clock port.

IF ASKED All attackers are Linux namespaces on one host; no real network or host clock was touched. C1 drew 60–75 % blackhole levels.

[illegible]

ETSI RFC 7394 cl. 3.2 [1] | ETSI TR 104 106 V3.0.0 T-SPLANE-05..04 [2] | ETSI TS 104 105 V7.0.0 O-RAN Security Test Specifications [3] | TMS926 sec. 4.1 [10]

ANSWER None of the eight threat classes was invented. Seven follow a published threat or test; C3 is our own instance of the RFC 7384 spoofing class, with no published test of that exact form.

IF ASKED C3 wording is bounded: we did not find a published test of that exact field combination in these sources. That is not proof none exists.

Measured ptp6 logs of RU1–RU3, 12 replicates per scenario (36 RU logs each), extracted_16bin_ptp6_logs | Impact classes: RFC 7594 Table 1 [1]; ETSI T-SG-PAN impacts [2]; TIMESAFE [30]

LIMIT Every daemon ran with `free_running: 1`: offsets were measured, never steered. Measured impact is protocol-level; time-error and service impact come from the cited literature.

IF ASKED free_running 1 means clocks were never steered, so the impact we measured is protocol-level. Time-error and service impact come from the cited literature.

Green et al., *TIME4SAC*, ACM TUPS 28(3), 2023, doi:10.1145/3775940 | Benati et al., *Colossus*, IEEE *OnCoRe*, 2024, doi:10.1109/ONCORE.2024.3447472
Digital Twin for O-RAN, IEEE *Communications Magazine*, 2024, doi:10.1109/MCOM.003.2400016 | OpenRAN Core, *Computer Networks*, 2023 | TSN Digital Twin, *EURASIP JIS*, 2023, doi:10.1186/153835-025-00213-5

Prior work establishes the threat, the AI tested direction and the digital twin direction. It does not resolve the operational decision between malicious timing behaviour and a benign timing fault presenting the same symptom.

IF ASKED The gap claim is bounded to the reviewed source set.

Federated Continual Learning for O-RAN Anomaly Detection, IEEE WCNC 2024 | Adversarial ML Threat Analysis in O-RAN, JNCAS 2024, doi:10.1016/j.jncas.2024.104099
O-RAN edge: Survey and Research Challenges, Computer Networks 2025 | Anomaly Detection for edge and 5G Threats, IEEE 2025

Gap addressed: the reviewed source set did not combine 3-plane benign lookalikes, standards-aware rules, a machine-learning arm, explicit abstention and a constant baseline within one held-out campaign.

SAY We designed the experiment so a learned model has to prove it adds value over a rule and over a constant before it gets authority.

Normative basis: IEEE 1588-2019, ITU-T G.8275.1, O-RAN WG4.CUS.0, O-RAN WG11 (5TSR TR 104 106 V3.0.0) | Prior-art boundary: TIMESAFE, Colossus, OpenRAN Gym, digital twin Emulation

Contribution: a standards-aware, controlled S-plane benchmark that tests whether a learned model adds decision value before it is granted authority over recovery.

SAY Our contribution is a standards-aware, controlled S-plane benchmark that tests whether a model adds decision value before it is trusted with recovery.

STANDARDS FOLLOWED

Telecom Standards and Profile Grounding

Normative basis: IEEE 1588-2019 | O-RAN WG4.CUS.0 | O-RAN WG4 Security Threat Modeling 2019 TR 104 106 | ITU-T G.8275.1
Timing performance context: ITU-T G.8273.2 | Frequency synchronization context: ITU-T G.8262 (BMC), not implemented here

AUTHORITY	WHAT IT DEFINES	HOW THIS PROJECT USES IT
IEEE 1588-2019	Legal PTP message fields, data sets, clock behaviour and leap-second handling.	ARM A checks field legality, sequence continuity and timePropertiesDS consistency.
O-RAN.WG4.CUS.0	Open Fronthaul CU/5-plane architecture and synchronization configurations.	Defines the O-DU-O-RU S-plane boundary used by the experiment.
O-RAN WG11	Open Fronthaul threat model, including malicious timing-source behaviour.	Supplies the threat identifiers that label the attack scenarios.
ITU-T G.8275.1	Telecom PTP profile for phase and time distribution with full timing support.	Flies domain, message rates, transport and the alternate BMCA used by the testbed.
ITU-T G.8273.2	Timing characteristics and limits for telecom boundary clocks.	Provides performance context for interpreting timing-quality evidence.

Scope disclosure

The testbed exercised PTP only; SyncE is a physical-layer function and was not implemented. ITU-T recommendations are paywalled, so several values were confirmed against authoritative secondary sources, as recorded in the project workbooks.

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STANDARDS

Telecom standards and profile grounding

SHOWS IEEE 1588-2019, O-RAN.WG4.CUS.0, O-RAN WG11 (ETSI TR 104 106), ITU-T G.8275.1 and G.8273.2: what each defines and how we used it.

SAY Every legality check traces to a standard, not an arbitrary threshold.

IF ASKED SyncE (G.8262) is context only; it is a physical-layer function and was not implemented.

EXPERIMENTAL CONFIGURATION

Open Fronthaul Timing Testbed Architecture

Images v4.0.0 released 2022-09-06, reference implementation of IEEE 1588-2019 and ITU-T G.8275.1 | Linux network namespaces, veth pairs and software bridges | Tscmap packet capture

Six genuine tscmap 4.0 daemons in isolated Linux network namespaces across two bridged segments. All protocol behaviour — BMCA election, servo control, message cadence — is executed by real Daemons, not simulated.

GM-A
generator

GM-B
backup OPI

BC
boundary clock

RUI1
radio unit

RUI2
radio unit

RUI3
radio unit

CAPTURE AND TIMETRY

Tscmap on both bridges during each run (raw PCAPs not kept in the packaged archive)

ptcpd servo logs from all six daemons

pmc management queries per run (returned no data; see KPI sources)

context.json — ground truth label plus provisioned operator context

ITU-T G.8275.1 profile as configured

domain 24 | priority1 128 | Sync 16/s (loginterval -4) | Announce 8/s (loginterval -3)

Evidence boundary

No independent clock-error instrument was used; architectural impact is inferred from protocol and servo telemetry, not measured against a reference clock.

Paths are protected by live packet path mechanisms — resampling, dropped frames, replayed captures and link or queue manipulation — never by editing runs in a dataset

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TESTBED

Open Fronthaul timing testbed architecture

SHOWS Six real linuxptp 4.0 daemons in namespaces on brUP and brDN. Capture, servo logs, pmc and context.json per run. Profile values.

SAY Everything protocol-related — election, servo, cadence — is executed by real daemons. Faults are produced on the live packet path, never by editing a dataset.

IF ASKED No independent clock-error instrument was used, so no conformance claim against G.8273.2 or G.8271.1 is made.

CONFIGURATION AND PLATFORM LIMITS

Timing Configuration and Verified Platform Constraints

ITU-T G.8275.1 telecom profile values cross-checked against Cisco and Nokia profile documentation and tscmap config/G.8275.1.cfg | Constraints confirmed by direct query of the running system

PARAMETER	CONFIGURED VALUE	BASIS
Profile	ITU-T G.8275.1 telecom profile	Full timing support from the network
domainNumber	24	G.8275.1 permitted range 24-43
priority1	128 (fixed)	G.8275.1 fixes the value and the alternate BMCA excludes it from comparison
Sync rate	16 / s (loginterval -4)	G.8275.1
Announce rate	8 / s (loginterval -3)	G.8275.1
BMCA	dataset_comparison G.8275.x	Alternate BMCA, not the IEEE 1588 default
Transport	L2 multicast 01:1B:19:00:00:00	Forwardable address, permitted by G.8275.1 (Ethernet default)
Timestamping	software	Platform limit — see constraints

Effect on scope

These constraints set the ceiling on every result that follows: five of the sixteen catalogued fault classes cannot be produced here, and no claim is made about absolute time-error magnitude.

Every configured value was read back from the running system and traced to the governing standard or, where the normative text is paywalled, to an authoritative secondary source.

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TESTBED

Timing configuration and verified platform constraints

SHOWS Each configured value (domain 24, priority1 128, Sync 16/s, Announce 8/s, alternate BMCA, L2 multicast) with its basis, plus four platform limits.

SAY These limits set the ceiling on every result: five of sixteen fault classes cannot be produced here, and we make no claim about absolute time-error size.

IF ASKED Values were read back from the running system, not restated from intent.

PROCESSING PIPELINE

Data Flow from Packet Capture to Verdict

Deep extractor: 56 fields per PTP frame | Field definitions from IEEE 1588-2019 message and data-net clauses | Conformance flags derived from ITU-T G.8275.1 profile constraints

01
The wire

Real PTP frames generated by six tscmap daemons on two bridged segments

02
Capture

Tscmap captures PTP frames on brUP and brDN; raw PCAPs not kept in the archive

03
Deep extract

56 protocol and conformance fields per packet; fields that cannot be observed are written EMPTY

04
Decision rule

Standards legality plus provisioned operator context, applied as ordered clauses

05
Verdict

ATTACK - BENIGN - UNKNOWN, with the triggering clause recorded

INTEGRITY RULE CARRIED THROUGH THE WHOLE PIPELINE

A value that cannot be measured is written EMPTY — never zero.

A zero offset reads as perfect synchronisation. Its earlier false writing stops for unobserved fields; any consequentially reported a blind node as healthy. Nothing is bound as healthy without positive proof of observation, and this rule is applied at extraction, at feature construction and at scoring.

The same extracted evidence feeds both decision areas, as the rule-based and machine-learning comparison is performed on identical extracted runs rather than on separately prepared datasets.

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PIPELINE

Data flow from packet capture to verdict

SHOWS Wire, capture, deep extract (56 fields), decision rule, verdict with the triggering clause. The integrity rule: unmeasured is EMPTY, never zero.

SAY A zero offset reads as perfect sync. An earlier build wrote zero for missing data and reported a blind node as healthy, so we made EMPTY-not-zero a rule everywhere.

IF ASKED Both arms are fed from the same executed runs, not separately prepared datasets.

KPI SOURCES

KPIs Are Measured From the Protocol, Not a Model

IEEE 1588-2019 message formats and data sets [1] | tscmap pcapd and pmc [14] | Extractor ptp, other, network.py; logs extractor_14dawn_ptcpd_logs_pmc_log.sh

01 Wire capture tscmap on brUP / brDN via ptp_dmap_vnetmap 56 fields per PTP message: header, flags, timestampoffset, announceProperties, sequenceid, announce data set, G.8275.1 conformance flags Used by the rule (20M A)	02 Daemon servo log ptcpd on output on each node master offset (ns), servo status, frequency adjustment (ns/s), path delay (ns), port-state transitions Used by the ML model (20M B)	03 Management queries pmc GET on 7 data sets per node, every second Intended: CURRENT, PRESENT, TIME, PROPERTIES, PORT data sets and port statistics Returned no data in any run	04 Operator context context.json written per run Provisioned allow-list, expected boundary clock, time-uncertainty statistics, ground-truth label Configuration, not measurement
--	--	--	--

Example servo log, baseline run 1, RUI1 ptpd[1517.414]: Master offset: 1291.98 Freq: -1547 path delay: 5313

ANALYSIS

No model generates the KPIs. They are read from PTP itself, defined by IEEE 1588-2019 and ITU-T G.8275.1. The ML model consumes six servo-derived features; it does not produce KPIs.

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KPIs

NEW

KPIs are measured from the protocol, not a model

SHOWS Four KPI sources: wire capture (rule), servo log (ML), pmc management queries (returned nothing), operator context (configuration).

SAY No model generates the KPIs. They are read from PTP itself, as defined by IEEE 1588 and G.8275.1. The ML model consumes six servo-derived features; it does not produce KPIs.

IF ASKED pmc: 40,320 unanswered requests, probably pmc without -d 24 (not re-run). RAN KPIs over E2 do not exist at the O-RU.

O-RAN S-Plane Timing Security · project guide

23

21 KPIS NEW

KPI classes and the recovery decision each drives

SHOWS Seven KPI classes (legality, identity, continuity, time properties, operator context, timing quality, not observable) mapped to ATTACK → ISOLATE, BENIGN → HOLDOVER, UNKNOWN → ESCALATE.

SAY Legality, identity, continuity and time-property KPIs decide ATTACK; context separates planned from unplanned; timing quality only grades severity.

IF ASKED Actions are recommended, not executed, in the 168-run campaign.

22 DATASET

Experimental dataset composition

SHOWS 168 runs, 14 scenarios, 12 replicates, 1,451,909 packet records, 56 fields.

What one run contains. Records per run vary with the fault (C1 $\approx$ 2,360; baseline $\approx$ 6,021; A5 $\approx$ 38,319).

SAY The variation in record count is itself evidence: interception starves the link, flooding inflates it.

IF ASKED Raw PCAPs (~222 MB) were captured but not kept in the packaged archive; the extracted CSVs, verdicts, logs and scripts were.

23 DATASET NEW

Where every dataset came from and how it is used

SHOWS The 168-run campaign (primary evidence), testbed configs (reference), TIMESAFE captures (reference only), the 13 Sep pilot (development), simulator output (not evidence).

SAY The evaluated dataset was generated by us on a reproducible testbed. The only external data, TIMESAFE, is public and cited.

IF ASKED August results that used TIMESAFE were withdrawn on 17 Sep.

24 SCENARIOS

Fault injection mechanisms on the live packet path

SHOWS Four mechanisms: extra real daemons (A1, A8), crafted frames (A2, C2, C3), capture and replay (A3), link and queue manipulation (A5, C1, B3, B7).

SAY An attack and its benign twin often share the same mechanism. Killing a grandmaster is identical in B2 and the unplanned failover; only the context differs.

25 SCENARIOS

Attack scenarios 1 of 2: A1, A2, A3, A5

SHOWS Injected condition, observable effect, expected verdict and result: 12/12 for each.

SAY All four were caught by the pre-frozen base rule at 12/12. Detection rests on legality and allow-list membership, not on how large the timing error is.

FAULT TAXONOMY - ATTACK SCENARIOS 2 OF 2				
Attack Scenarios: Relay, Interception and Field Abuse				
G-RAN WGS1 security test identifiers L1, L3, L5 and 24.2, L2 2750 TO 504 100 IEEE 1588-2019 implementation, IEEE802.15.4-2015 and loop-second handling 50-4 ETSI TS 104 106 V5.0.0				
SCENARIO	INJECTED CONDITION	OBSERVABLE EFFECT	EXPECTED	RESULT
A8 Relay boundary clock TSAP-46-02	A two-port p2p4 instance slaves to the legitimate grandmaster upstream and re-advertises downstream with registered maintenance incremented.	An unauthorized clock is inserted into the timing path while still relaying genuine time, so content alone appears valid.	ATTACK	12/12 correct
C1 Covertive interception and removal TSAP-46-04	The boundary clock's downstream port is blackholed for a sustained fraction of the observation window, withholding sync and Announce.	Radio units receive messages below the rate the boundary clock itself declares, delivered to declared ratio falls toward 0.57.	ATTACK	11/12 correct
C2 Malformed / faked frames WGS1 24.2.1.2	Frames injected with version 07P-3, a reserved message type, and a message length shorter than the protocol permits.	Frames are structurally illegal under IEEE 1588-2019 regardless of their timing content.	ATTACK	12/12 correct
C3 Whole-second field abuse IEEE 1588-2019 50.2.4	Announce frames carrying loop4s asserted, currentOffset failed to 0 against a true value of 37, and timestamp flags set-ign.	Time properties become internally inconsistent, which would shift the applied UTC offset by a whole second.	ATTACK	12/12 correct
Note The pre-frozen base rule returned BENIGN on C1, C2 and C3 at 0/12 each. These three coverage gaps were closed by a disclosed post-campaign rule revision, evaluated on the same operators and reported separately from the pre-frozen result.				

26 SCENARIOS

Attack scenarios 2 of 2: A8, C1, C2, C3

SHOWS A8 12/12, C1 11/12, C2 12/12, C3 12/12 with v3.

SAY The pre-frozen base rule returned BENIGN on C1, C2 and C3 at 0/12. These gaps were closed by a disclosed revision, reported separately.

IF ASKED The C1 miss is one replicate at a 60 % blackhole; see slide 36.

FAULT TAXONOMY - BENIGN SCENARIOS				
Benign Fault and Operational-Change Scenarios				
ITU-T G.8275.1 docClass label and timestamped semantics IEEE 1588-2019 50-4 re-election behavior Provisioned operator control: allow list, maintenance window state, expected boundary-clock identity				
SCENARIO	INJECTED CONDITION	OBSERVABLE EFFECT	EXPECTED	RESULT
baseline Healthy control —	All six domains nominal: grandmaster A elected, allowed through the boundary clock to three radio units.	Stable offset and path delay; message cadence matches the declared profile rates.	BENIGN	12/12 correct
B2 Planned grandmaster changeover MPL 6.8.075.1	Grandmaster A withdrawn during an open maintenance window; grandmaster B, which is on the provisioned allow-list, is elected.	A grandmaster change identical in packet format to A1 — the distinguishing evidence is allow-list membership plus maintenance state.	BENIGN	12/12 correct
B3 Packet delay variation and congestion MPL 6.8.081	A 1 Mbit Tbf bottleneck on the boundary-clock to radio unit path carrying competing non-P2P background traffic.	Real operating delay: Sync packet delay variation rises from 0.07 ms to between 5.6 and 87 ms, with no packet loss and no protocol anomaly.	BENIGN	11/12 correct
B7 Topology reconfiguration IEEE 1588-2019	A radio-unit link is lossened and re-formed during an open maintenance window.	discontinuous and measured path delay change consistently with the new topology, with no illegal field values.	BENIGN	12/12 correct
B_bc_repl Provisioned boundary-clock replacement MPL 6.8.075.1	A provisioned replacement boundary clock is swapped in during maintenance. Its identity is recorded in the operator control.	A new clock identity appears in the timing path replacing the genuine grandmaster — structurally identical to the A8 relay relay.	BENIGN	0/12 open value
Note Four of the benign scenarios were classified correctly. The boundary-clock replacement in the single system, below, the decisive control carries one expected boundary-clock identity, so a legitimate replacement is read as an unauthorized relay.				

27 SCENARIOS

Benign fault and operational-change scenarios

SHOWS baseline 12/12, B2 12/12, B3 11/12, B7 12/12, B_bc_replacement 0/12.

SAY Four of five benign scenarios are correct. The boundary-clock replacement is the single systematic failure.

IF ASKED The B3 miss is an UNKNOWN, a cautious abstention, not a false alarm.

FAULT TAXONOMY - AMBIGUOUS CASE AND COVERAGE				
Ambiguous Scenario and Catalogue Coverage				
System-class System fault catalogue maintained in the project parameter matrix (344 parameters × 38 fault classes) Threat identifiers from ETSI TS 104 106 V5.0.0 Coverage assessed against verified platform constraints				
B-unplanned-failover - DISCREPANT AMBIGUOUS Grandmaster A falls with no maintenance window open; provisioned grandmaster B is elected. From packet evidence alone this is indistinguishable from an attack that superseded the grandmaster. Both produce parameter evidence for a legitimate failure. The decision relies on context (catalogue and maintenance window timing) and a binary answer.			EXPECTED UNKNOWN 12 / 12 correct	
SEVENTEEN-CLASS CATALOGUE COVERAGE				
A1 new TESTED Illegal grandmaster / BMCA spoof	A2 new TESTED Sync follow-up spoofing	A3 new TESTED Invalid	A4 new TESTED Sync threat attack	not run
A5 new TESTED Out-of-P2P spoofing	A6 new TESTED Clock spoofing	A7 new TESTED Clock jamming	A8 new TESTED Relay boundary clock	TESTED
B1 new TESTED Clock holdover	B2 new TESTED Planned GM changeover	B3 new TESTED P2P / congestion	B4 new TESTED SyncC / SSC degradation	TESTED
B5 new TESTED Static path asymmetry	B6 new TESTED Clock drift	B7 new TESTED Topology reconfiguration	B8 new TESTED Measurement noise floor	not run
Eight of sixteen catalogued classes were executed. They require physical timing hardware and cannot be produced here. There have software detection logic only, where physical impact would need hardware to measure. SW* denotes detection logic without physical-impact measurement.				

28 SCENARIOS

Ambiguous scenario and catalogue coverage

SHOWS B_unplanned_failover returns UNKNOWN 12/12. The 16-class catalogue: 8 executed, 3 software-detection only, 5 hardware-only.

SAY When the packets cannot reveal intent, the rule escalates instead of guessing. A BENIGN call there would be unsafe and an ATTACK call a false alarm.

CLASSIFICATION METHOD	
Classification Logic and Verdict Assignment	
IEEE 1588-2019 field legality, data set consistency and response conformity ITU-T G.8275.1 profile constraints and docClass label Provisioned operator control supplied per run in control panel	
BENIGN SIGNATURE Degradation that is self-consistent with a declared, standards-compliant state change: docClass 6 to 7 for a grandmaster entering holdover within specification, timestamped changing consistently with a new topology, or delay variation rising without any field becoming illegal. The declaration and the observation agree.	ATTACK SIGNATURE Evidence that is illegal or unauthorized: forged Announce fields, non-monotonic sequence identifiers, structurally invalid frames, a delivered message rate below the rate the source itself declares, or a superior clock whose identity is outside the provisioned allow-list. The declaration and the observation disagree.
VERDICT ASSIGNMENT AND RECOMMENDED RESPONSE	
ATTACK Protocol inconsistent or unauthorized evidence present ISOLATE	BENIGN Only provisioned clocks, no legality violation TOLERATE / HOLDOVER
UNKNOWN Packet evidence cannot distinguish intent ESCALATE	
Recommended response labels only. No corrective action was executed in this 168-run campaign.	
The worked discriminator: a grandmaster suddenly changes. It could be B2, a planned failover; or A1, BMCA spoofing. The decisive test is whether the new clock identity appears on the provisioned allow-list and whether a maintenance window is open. Both scenarios were classified correctly at 12/12.	
On the deliberately ambiguous unplanned failover the rule returned UNKNOWN on 12 of 12 runs. A BENIGN call there would be an unsafe over-claim and an ATTACK call a false alarm; both are treated as follows.	

29 METHOD

Classification logic and verdict assignment

SHOWS Benign signature (declaration and observation agree) versus attack signature (they disagree), and the verdict-to-response mapping.

SAY The worked discriminator: a grandmaster suddenly changes. The decisive test is whether the new identity is on the allow-list and whether maintenance is open.

IF ASKED Response labels only; no isolation, holdover or escalation was executed in this campaign.

EXPERIMENT DESIGN	
Campaign Design and Reproducibility Controls	
Decision rule 504-256 hashed before execution (PROCEED) Per replicate randomisation seeded from the replicate number Pre-registered expected verdicts Full run inventory in FOUNDATION, V3.0.0	
168 live on-wire runs	14 scenarios
12 randomised replicates each	3 rules scored on identical captures
Rule frozen before execution	The base decision rule was 504-256 hashed before any run started, so it could not be tuned to the outcome. The hash is recorded in the packaged archive.
Revision history disclosed	Rule v3 fixed v2 false positives seen on upstream captures and was re-hashed before scoring. v2, frozen before the campaign began, gives identical verdicts on all 168 runs.
Pre-registered expectations	Each scenario's expected verdict was written down in advance. Sensitivity, specificity and abstention are scored on separate axes rather than pooled into one figure.
Per-replicate randomisation	Attacker identity, priority, burst size, inter-frame gap and onset are drawn reproducibly from the replicate number, so repeats test generalization rather than repetition.
Per-scenario reporting	Per-scenario recall is the primary metric. Pooled figures move with how many runs of each scenario exist, so they are labelled descriptive only.
Design limit	The three C-series scenarios draw from only four to five distinct randomized parameter values, so twelve replicates are twelve genuine captures but not twelve independent attack designs.

30 METHOD

Campaign design and reproducibility controls

SHOWS Rule hashed before execution, revision history disclosed, pre-registered expectations, per-replicate seeded randomisation, per-scenario reporting.

SAY The rule could not be tuned to the outcome. v2, frozen before the campaign, gives identical verdicts to v3 on all 168 runs.

IF ASKED C-series scenarios draw from only four to five distinct parameter values, so twelve captures are not twelve independent designs.

EXPERIMENTAL INTEGRITY

Pipeline Audit and Corrective Reruns

Independent audit of the campaign apparatus, 2024-09-26 | Detect evidence and corrective actions necessary to **FIXED, CORRECTING** | All affected runs re-executed before any result in this deck was computed

A cross-check audit of the campaign apparatus identified four defects. Two of them had produced invalid data. Every affected run was re-executed, and the results reported later in this deck are computed only from the corrected campaign.

Thirty-six replicates were three captures coded twelve times

A benign scenario never injected anything

A detector produced false positives on upstream captures

The evaluator overstated and under-checked

Proof: one distinct 30k-25k across twelve supported replicates, and completion in under one second under a generous one takes about 48 seconds.

Proof: B3 packet delay variation was statistically identical to baseline and to the unpatched control node. Result with a full bottleneck and competing traffic.

Effect: it turned a benign follower into an ATTACK and destroyed a correct abstention. Rule v3 adds a minimum-frame floor and a persistence requirement.

Replaced with a per-scenario evaluator applying macro averages, per-run additive only verification and an explicit replicable verdict.

FIXED - re-run

FIXED - rethink

FIXED - v3

REPLACED

Repeating actions fixed to one run capture is treated as part of the experimental result. Two of the four had fabricated data, both were corrected and re-executed, and the withdrawn figures are not reported anywhere in this deck.

31 METHOD

Pipeline audit and corrective reruns

SHOWS The four defects found on 20 Sep, two of which fabricated data, and how each was fixed.

SAY Disclosing defects in our own pipeline is part of the result. Withdrawn figures appear nowhere in this deck.

IF ASKED The earlier 132-run campaign is withdrawn because the B3 defect affected it too.

CLASSIFICATION RESULTS

Scenario-Level Classification Results

568 live runs, 14 scenarios, 12 replicates each | Wilson score 95% confidence intervals | Expected verdicts pre-registered before execution | Source: EVALUATION_M4.json

SCENARIO	EXPECTED	PRE-FROZEN BASE RULE	REVISED RULE v3	95% CI (v3)
A1 niger grandmaster	ATTACK	12/12	12/12	0.757-1.000
A2 sync spoofing	ATTACK	12/12	12/12	0.757-1.000
A3 replay	ATTACK	12/12	12/12	0.757-1.000
A4 out flooding	ATTACK	12/12	12/12	0.757-1.000
A8 niger boundary clock	ATTACK	12/12	12/12	0.757-1.000
C1 selective interception	ATTACK	0/12	11/12	0.646-0.985 gap closed
C2 malformed frames	ATTACK	0/12	12/12	0.757-1.000 gap closed
C3 whole-second abuse	ATTACK	0/12	12/12	0.757-1.000 gap closed
benign	BENIGN	12/12	12/12	0.757-1.000
B2 planned GM challenge	BENIGN	12/12	12/12	0.757-1.000
B3 congestion	BENIGN	11/12	11/12	0.646-0.985 1 observation
B7 topology reconfiguration	BENIGN	12/12	12/12	0.757-1.000
B_rc_replacement	0/12	0/12	0.000-0.242 open failure	
B_unmodified_benign	UNKNOWN	12/12	0.757-1.000	

The scenario result is the primary metric. Three coverage gaps — interception, malformed frames and whole-second abuse — moved from 0/12 to 11/12, 12/12 and 12/12 with no change in benign performance. One benign scenario remains an open failure.

32 RESULTS

Scenario-level classification results

SHOWS All 14 scenarios, base rule versus v3, with Wilson 95 % intervals.

SAY Three coverage gaps moved from 0/12 to 11/12, 12/12 and 12/12 with no change in benign performance. One benign scenario remains an open failure.

IF ASKED Intervals are wide by construction at n = 12 (12/12 gives 0.757–1.000).

CLASSIFICATION RESULTS

Aggregate Classification Performance

Macro averages over per-scenario correctness rates, 568 live runs | Attack sensitivity over 8 attack scenarios, specificity over 5 benign scenarios, abstention over 1 ambiguous scenario | Source: EVALUATION_M4.json

0.990

Macro attack sensitivity

revised rule v3 - pre-frozen base rule 0.625 on the same captures

0.783

Macro benign specificity

identical for both rules, so the revision introduced no regression

1.000

Abstention correctness

ambiguous unpatched behavior returned UNKNOWN on 12 of 12

0.875

Fault attribution accuracy

84 of 96 attack runs had the specific fault correctly named

WHAT IMPROVED

The revised rule closed three coverage gaps on the same captures: selective interception 0/12 to 11/12, malformed frames 0/12 to 12/12, and whole-second abuse 0/12 to 12/12. Benign performance is unchanged, and per-run verification confirms zero regressions — no run that the base rule called ATTACK was downgraded.

WHAT SPECIFICITY IS 0.783 AND NOT 1.000

Two benign scenarios miss. B_rc_replacement produces 12 false ATTACK calls and is the dominant cause. B3 congestion contributes one UNKNOWN. Excluding the boundary-clock replacement, benign classification is 47 of 48, or 0.979 — so the loss is one identified systematic gap plus one cautious abstention, not diffuse error.

Reporting integrity

The 0.990 figure belongs to the post-observation revised rule. The pre-frozen base rule scored 0.625 on the same captures. Both are shown as the improvement is not mistaken for independent pre-registered validation.

Real-time impact: among attack detection with one identified and explainable benign failure, no benign abstention on the previously undecidable case, and the exact fault named on 84 of 96 attack runs.

33 RESULTS

Aggregate classification performance

SHOWS Sensitivity 0.990 (base 0.625), specificity 0.783, abstention 1.000, attribution 0.875 (84/96).

SAY Strong attack detection, one identified and explainable benign failure, and an honest abstention on the undecidable case.

IF ASKED Attribution is fully correct on 4 of 8 scenarios; the 12 misses are rogue-clock runs labelled as replay and whole-second runs caught by the interception clause (see Part III). The closing line was corrected on 5 Oct from an unsupported 'seven of eight scenarios' to '84 of 96 attack runs'.

RULE-BASED AND ML COMPARISON

Rule-Based and Machine-Learning Classification Compared

Worklet objective 5 requires comparison against baseline fault management | Session object split: trained on replicates 1-8, tested on replicates 9-12 (56 held-out runs) | Hyperparameters and split fixed before scoring

ARM A - RULE-BASED

Frozen decision rule

Team 5 submitted pre-registered baseline pre-processed operator control. No training. 30k-25k frames. No coverage. 0% false detection per capture metric.

ARM B - MACHINE LEARNING

Random Forest + Isolation Forest

97 runs, 48k+ 16-bit frames, used 28k+ with an isolation forest novelty test for anomalies. Inputs: ping-pong telemetry only — offset, path delay and delay variation.

CONTROL - ALWAYS-BENIGN

Constant predictor

Returns BENIGN for every run. Included so that any apparent ML fault can be isolated against the trivial baseline (rather than assumed).

METRIC ON 56 HELD-OUT RUNS

	ARM A - RULE	ARM B - ML	ALWAYS-BENIGN
Exact accuracy	51/56 (0.911)	24/56 (0.429)	20/56 (0.357)
Attack sensitivity	0.969 (31/32)	0.156 (5/32)	0.000 (0/32)
Benign specificity	0.800 (16/20)	0.950 (19/20)	1.000 (20/20)
Correct abstention	1.000 (14/14)	0.000 (0/4)	0.000 (0/4)

ARM B IS NOT DETECTABLE FROM THE CONSTANT PREDICTOR

ARM B means BENIGN on 47 of 48 runs and improves on the constant predictor by two runs. A team 5 pre-registered baseline pre-processed operator control. No training. 30k-25k frames. No coverage. 0% false detection per capture metric. No training. 30k-25k frames. No coverage. 0% false detection per capture metric.

NO MEASURED COMBINATION EXCEEDS THE RULE ARM

Combining the arms was evaluated rather than assumed. Logical OR machine 0.875 rule for rule with ML, without 0.855, and combine with domain 0.875. Actual false data A at 0.911, so no combination is recommended on this evidence.

Scope condition

ARM A operated on 6 of 28 configured features because DNS, sync, oscillator and protocol-legality telemetry do not exist in this testbed. This is a result about the available observability, not a general verdict on machine learning.

34 RESULTS

Rule-based and machine-learning classification compared

SHOWS On 56 held-out runs: rule 51/56, ML 24/56, always-BENIGN 20/56. ML not distinguishable from the constant (McNemar p = 0.219). No combination beats the rule.

SAY This answers worklet objective 5. On the available telemetry, the transparent rule is the right operational choice.

IF ASKED ML used 6 of 28 features. This is a result about observability, not a verdict on machine learning in general.

RESULT INTERPRETATION

Observability Constraints Behind the ML Result

Feature availability derived from observed telemetry server presence | Schema details and synthetic panel fields normalized to facts before fitting | Inquiry-based feature importances from the fitted Random Forest

The machine-learning arm underperformed on what this testbed can observe. Three measured constraints bear on the result: an equal-input comparison was not run, so method suitability level is untested.

01

Only 6 of 28 configured features carried data

QNS) status, sendoff quality, oscillator error and every protocol-legality feature are absent from global server logs. The model could rely only on offset, path delay and delay-variation magnitude — and one of the six, holdover rates, had zero importance, so few features carried all the signal.

02

Two attacks starve the very telemetry the model needs

Seven samples average 88.0 per baseline run, falling to 44.5 for selective interception and 28.0 for whole-second abuse. An attack that suppresses timing messages also suppresses the evidence a telemetry-based detector depends on. The rule arm is unaffected because it reads packet legality and rate deficit directly.

03

Importances indicate testbed-servo artefact risk

Ranked importances are offset, abs, max 0.248, path, delay, mean 0.262, offset, abs 0.268, path, abs 0.139 and offset, max 0.128 — all timing magnitude and dispersion, none protocol semantics. With a single testbed, domain shift cannot be separated from intrinsic class overlap.

Baseline-based conclusion: on this telemetry the rule arm is the correct operational choice. Variation alone does not explain the result — ARM A also fails on attacks where server telemetry is intact — so feature poverty and class overlap are the primary causes. A fair re-test requires either management-plane telemetry, not a different classifier.

This is an argument for extending a standard expert rule test and for improving observability before granting a learned model authority over recovery. It is not a claim that machine learning is inherently unsuitable to detect timing events.

35 RESULTS

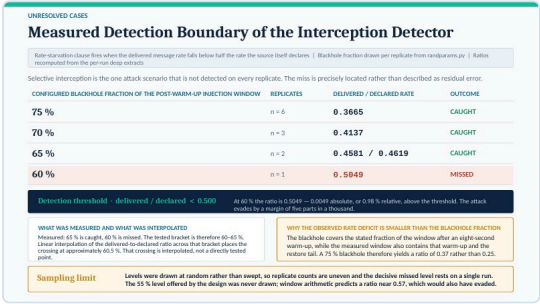
Observability constraints behind the ML result

SHOWS Only 6 of 28 features carried data; two attacks starve servo telemetry (88.0 → 44.5 and 28.0 samples); importances are all timing magnitude.

SAY Starvation alone does not explain it — the model also fails where telemetry is intact. Feature poverty and class overlap are the main causes. A fair re-test needs richer management-plane telemetry, not a different classifier.

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36 LIMITS

Measured detection boundary of the interception detector

SHOWS Blackhole 75 %, 70 %, 65 % caught; 60 % missed (ratio 0.5049 against the 0.500 threshold).

SAY The miss is located precisely: it evades by five parts in a thousand. Interpolating puts the crossing near 60.5 %, but that point was not directly tested.

IF ASKED The 55 % level was never drawn and would also have evaded (predicted ratio ≈0.57), so recall over the whole design space is weaker than 11/12 suggests.



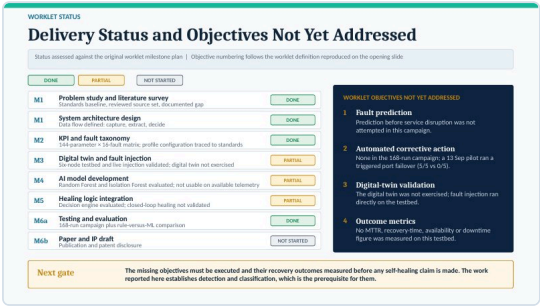
37 LIMITS

Experimentally verified limitations

SHOWS Five limitations, each with evidence, technical reason and implication.

SAY Each limitation is a finding with a located cause. Four of five have a defined route to resolution.

IF ASKED B_bc_replacement: the replacement identity is provisioned and v3 reads it, but the base rule's rogue-BC clause reads only the primary identity, and additive-only v3 cannot overturn it.



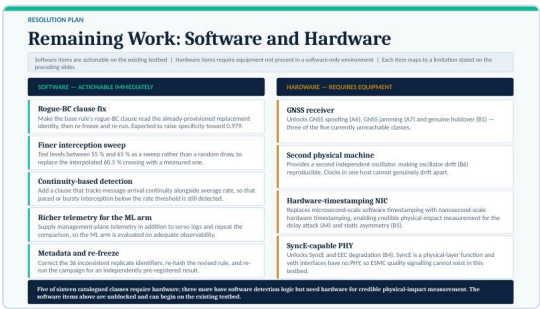
38 STATUS

Delivery status and objectives not yet addressed

SHOWS M1, M2 and M6a done; M3, M4 and M5 partial; M6b (paper and IP) not started. Prediction, automated action, twin validation and outcome metrics are not addressed.

SAY The work establishes detection and classification, which is the prerequisite for self-healing. No self-healing claim is made yet.

IF ASKED The only executed action was the 13 Sep pilot port failover (5/5 vs 0/5); its independent validation S15 was not established.

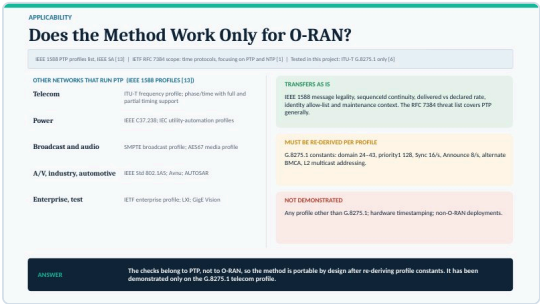


39 STATUS

Remaining work: software and hardware

SHOWS Software now: fix the rogue-BC clause, sweep interception levels, add continuity detection, richer ML telemetry, fix metadata and re-freeze. Hardware: GNSS receiver, second machine, HW-timestamping NIC, SyncE PHY.

SAY The software items are unblocked and can start on the existing testbed. A GNSS receiver is the single highest-value hardware item.



40 APPLICABILITY NEW

Does the method work only for O-RAN?

SHOWS PTP is used in telecom, power, broadcast, automotive and industry. What transfers as is, what must be re-derived per profile, and what is not demonstrated.

SAY The checks belong to PTP, not to O-RAN, so the method is portable by design after re-deriving profile constants. It has been shown only on G.8275.1.

TECHNICAL SUMMARY

Open Fronthaul S-Plane Timing Security

1 Scope investigated

Discrimination of malicious timing behaviour from benign timing faults on the Open Fronthaul S-plane, between O-DU and O-RU.

2 Implementation

Six Inuapp 4.0 detectors under the ITU-T G.8275.1 telecom profile, across two bridged segments, with faults produced on the live packet path.

3 Dataset produced

168 live test sessions (4 scenarios at 12 randomised replicates each, 1,431,909 packet records extracted at 16 fields per packet, with underwrite fields reserved EAPY).

4 Taxonomy evaluated

Sign attack scenarios, five benign scenarios and one deliberately ambiguous case, drawn from a sixteen-class catalogue of which eight were executable in software.

5 Classification performance

Malice attack sensitivity 0.990 for the revised rule, against 0.625 pre-campaign sensitivity (0.705, or 0.979 excluding the identified benign failure; abstention 12/12, abstention 0.87%).

6 Comparison outcome

On 56 held-out runs the rule arm reached 0.911 against 0.429 for the ML arm and 0.267 for a constant predictor; the ML arm is not statistically distinguishable from that control on the available telemetry.

7 Limits identified

One systematic benign failure with a defined fix, an interception evasion band near 60 %, and five fault classes requiring hardware.

8 Immediate next steps

Extend the pre-campaign requirement context and re-process: sweep interception levels, add continuity-based detection, re-run the ML arm with management-plane telemetry.

Thank you

Tannaya Kumar · Raghu Ram K · Murugesh Jeyaseeth Kumar

Status: S-plane detection and classification evaluated. Prediction, digital twin and automated healing not yet executed.

41 CLOSING

Technical summary

SHOWS Eight-point summary: scope, implementation, dataset, taxonomy, performance, comparison, limits, next steps.

SAY Status: S-plane detection and classification evaluated. Prediction, digital twin and automated healing not yet executed.

IF ASKED Most important caveat: 0.990 belongs to the rule revised after observing the campaign; the pre-frozen rule scored 0.625 on the same captures.

REFERENCES

References for the Review-Response Slides

Numbered as cited on these slides | Sources of the original slides remain on those slides | Verified on 3 October 2023

[1] T. Mizrahi, "Security Requirements of Time Protocols in Packet Switched Networks," IETF RFC 7584 (Informational), Oct. 2014.

[2] ETSI TR 104 104 V1.0.0 (2023-06), O-RAN Security Threat Modeling and Risk Assessment (O-RAN WG11).

[3] ETSI TS 104 104 V7.0.0 (2023-06), O-RAN Security Test Specifications.

[4] ETSI TS 103 182 V8.0.0 (2024-03), O-RAN Architecture Description.

[5] IEEE Std 1588-2019, Precision Clock Synchronization Protocol for Networked Measurement and Control Systems.

[6] ITU-T G.8275.1, Precision time protocol telecom profile for phase/time synchronization with full timing support from the network.

[7] G. Armitage, "O-RAN Fundamentals," ATIS WP5T3 2023, quoting O-RAN WG4 CUS.2 v06.00.

[8] S. Ruffini, "Sync for 5G: What is Needed," ATIS 2018, quoting 3GPP TS 38.133 cell phase synchronization accuracy.

[9] 3GPP TS 38.104 Rel-15, NR Base Station radio transmission and reception, clause 9.6.3.2 Time alignment error.

[10] I. Green et al., "TIMESAFE: Timing Information Monitoring and Security Assessment for Trusted Environment," ACM TOPS 28(3), 2023.

[11] ETSI TS 138 401 (3GPP TS 38.401) Rel-18, NG-RAN Architecture description.

[12] BSI (Federal Office for Information Security), 5G Risk Analysis: Radio Access Network, v1.2, 20 Feb 2023.

[13] IEEE SA, IEEE 1588 PTP profiles list, [sagepub, ieee.org/1588/ptp-profiles](https://www.ieee.org/1588/ptp-profiles).

[14] Inuapp project, ptp4j and ptpc documentation, inuapp.net/time.org.

42 CLOSING NEW

References

SHOWS References [1]–[14] cited on the new slides.

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PART III

The results

Every figure below comes from `EVALUATION_V4.json` and the per-run traceability workbook of the corrected 168-run campaign. The arithmetic is shown so you can rebuild any number on a whiteboard.

Per-scenario results

Per-scenario recall is the primary metric: for each scenario, how many of its 12 runs got the expected verdict.

SCENARIO	EXPECTED	PRE-FROZEN BASE RULE	REVISED RULE V3	95 % CI (V3)
A1 rogue grandmaster	ATTACK	12/12	12/12	0.757–1.000
A2 Sync spoofing	ATTACK	12/12	12/12	0.757–1.000
A3 replay	ATTACK	12/12	12/12	0.757–1.000
A5 DoS flooding	ATTACK	12/12	12/12	0.757–1.000
A8 rogue boundary clock	ATTACK	12/12	12/12	0.757–1.000
C1 interception	ATTACK	0/12	11/12	0.646–0.985
C2 malformed frames	ATTACK	0/12	12/12	0.757–1.000
C3 whole-second abuse	ATTACK	0/12	12/12	0.757–1.000
baseline	BENIGN	12/12	12/12	0.757–1.000
B2 planned GM changeover	BENIGN	12/12	12/12	0.757–1.000
B3 congestion	BENIGN	11/12	11/12 (1 UNKNOWN)	0.646–0.985
B7 topology change	BENIGN	12/12	12/12	0.757–1.000
B_bc_replacement	BENIGN	0/12	0/12 (open failure)	0.000–0.242
B_unplanned_failover	UNKNOWN	12/12	12/12	0.757–1.000

What "95 % CI" means here. With only 12 runs, even a perfect 12/12 cannot prove the true rate is 100 %. The Wilson interval says the true rate is plausibly anywhere from 0.757 to 1.000. Wide intervals are expected at this sample size, and saying so is the correct move.

The headline numbers, with the arithmetic

Each headline is a **macro average**: the average of the per-scenario rates, so every scenario counts equally.

Attack sensitivity: "of the attacks, how many did we call ATTACK?"

v3: $(1 + 1 + 1 + 1 + 1 + 11/12 + 1 + 1) / 8 = 7.9167 / 8 = \mathbf{0.990}$

Base rule: $(1 + 1 + 1 + 1 + 1 + 0 + 0 + 0) / 8 = 5 / 8 = \mathbf{0.625}$

Benign specificity: "of the honest faults, how many did we leave alone?"

baseline $1 + B2\ 1 + B3\ 11/12 + B7\ 1 + B_bc_replacement\ 0 = 3.9167 / 5 = \mathbf{0.783}$ (same for both rules, so no regression)

Without B_bc_replacement: 47 of 48 benign runs correct = **0.979**

Abstention: "on the undecidable case, did we escalate?"

B_unplanned_failover returned UNKNOWN on **12 of 12** runs = 1.000. This is scored on its own axis, not counted as a detection.

Attribution: "did we name the right attack?"

Exact fault named on **84 of 96** attack runs = 0.875 (Wilson 0.794–0.927).

Where the 12 attribution misses are (per-run workbook)

SCENARIO	DETECTED	EXACT FAULT NAMED	WHAT THE OTHERS WERE NAMED AS
A1 rogue GM	12/12	9/12	3 named as replay (A3)
A8 rogue BC	12/12	8/12	4 named as replay (A3)
C1 interception	11/12	11/12	1 missed (BENIGN)
C3 whole-second	12/12	8/12	4 caught by the interception clause
A2, A3, A5, C2	12/12 each	12/12 each	—

Corrected on 5 Oct. Slide 33's closing line used to say "correct fault attribution on seven of eight attack scenarios". The per-run data does not support that: attribution is fully correct on 4 of 8 scenarios (A2, A3, A5, C2). The line now reads "the exact fault named on 84 of 96 attack runs". If asked: *"Every attack was still detected; the misattributions are rogue-clock runs labelled as replay and whole-second runs caught by the interception clause."*

Why specificity is 0.783 and not 1.000

Two benign scenarios miss. **B_bc_replacement** produces 12 false ATTACK calls and is the main cause. **B3 congestion** produces one UNKNOWN, which is a cautious abstention, not a false alarm. The cause of B_bc_replacement is located exactly. The replacement clock's identity is provisioned (`expected_bc_identity_secondary`), and v3 reads it. But the base rule's rogue-BC clause (A8) reads only the primary identity, and v3 is additive-only, so it cannot overturn an ATTACK. The fix is to make that clause read the provisioned replacement, then re-freeze and re-run.

Rule versus machine learning

56 held-out runs: 14 scenarios $\times$ replicates 9–12. That is 32 attack runs, 20 benign runs and 4 ambiguous runs.

METRIC	ARM A · RULE	ARM B · ML	ALWAYS-BENIGN
Exact accuracy	51/56 (0.911)	24/56 (0.429)	20/56 (0.357)
Attack sensitivity	0.969 (31/32)	0.156 (5/32)	0.000 (0/32)
Benign specificity	0.800 (16/20)	0.950 (19/20)	1.000 (20/20)
Correct abstention	1.000 (4/4)	0.000 (0/4)	0.000 (0/4)

Check the totals. Rule: $31 + 16 + 4 = 51$. ML: $5 + 19 + 0 = 24$. Constant: $0 + 20 + 0 = 20$. The rule's 4 benign misses are the 4 held-out B_bc_replacement runs.

Is the ML model better than doing nothing?

The ML model said BENIGN on 47 of 56 runs, so it behaves almost like the constant predictor. The two disagree on only 6 runs: the model is right on 5 of them and the constant on 1. The **exact McNemar test** asks whether a 5-to-1 split could easily happen by chance if both were equally good. With 6 coin flips, a split at least that lopsided happens with probability $2 \times (1 + 6) / 64 = 0.219$. That is far above 0.05, so **the model is not distinguishable from the trivial baseline** at this sample size.

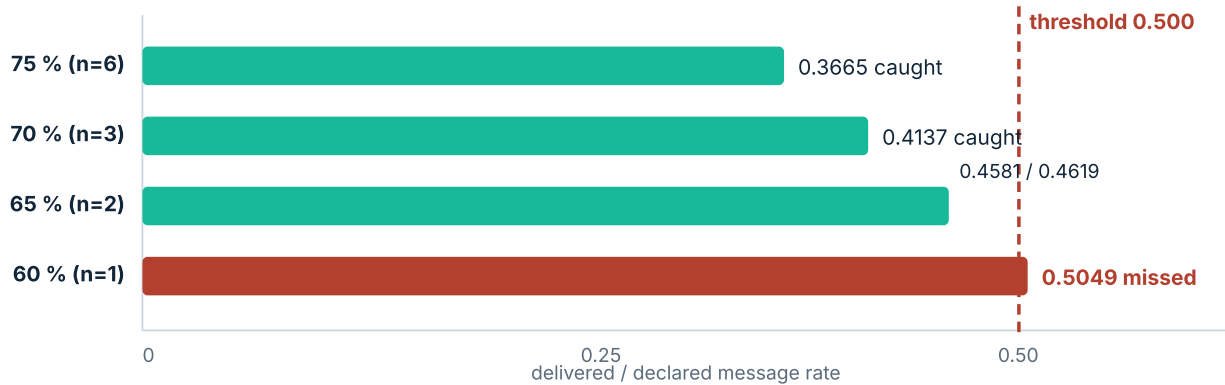
Combining them does not help either. Logical OR scores 0.893, rule-first with ML fallback 0.839, and consensus-else-abstain 0.429. All are below the rule alone at 0.911.

How to frame it. "On this telemetry the rule arm is the correct operational choice. That is a result about observability: the model saw 6 of 28 planned features. It is not a claim that machine learning cannot work for S-plane security. A fair re-test needs richer management-plane telemetry, not a different classifier."

Window-level figures exist separately (window accuracy 0.544, ROC-AUC 0.604 over 1,923 windows). They are per window, not per run, and should not be compared with the run counts above.

Where the detector stops working

The interception detector fires when the radio units receive fewer than **half** the messages the source declares it is sending. Here is what the 12 C1 runs showed.



Configured blackhole fraction (left) against the measured delivered-to-declared rate. Bars to the left of the red line are caught.

- At 60 % the ratio is 0.5049, which is 0.0049 above the threshold: the attack evades by five parts in a thousand.
- **Measured:** 65 % caught, 60 % missed. **Interpolated, not tested:** the crossing is near 60.5 %.
- Why a 75 % blackhole gives 0.37 and not 0.25: the blackhole covers 75 % of the window *after* an eight-second warm-up, but the measured window also contains the warm-up and the restore tail.
- The 55 % level was offered by the design but never drawn. Arithmetic predicts a ratio near 0.57, which would also evade. Recall over the full design space is therefore weaker than 11/12 implies, and the deck says so.

Limits, and what is not done

Five verified limitations

LIMITATION	EVIDENCE	ROUTE FORWARD
Planned BC replacement read as an attack	0/12	Make the rogue-BC clause read the provisioned replacement; re-freeze; re-run. Expected specificity near 0.979.
Interception below about 60 % evades	1 of 12 C1	Sweep 55–65 %; add a continuity-based clause.
Coherent GNSS spoofing is not establishable	testbed limit	Needs RF observables or an independent time anchor (hardware).
Held-out evidence is small; pmc empty	4 per scenario	Fix pmc (domain flag) and repeat the comparison with richer telemetry.
Replicate metadata inconsistent	36 C-series context files say rep 901	Correct before any fresh validation; the evaluation already uses the directory name.

Facts that limit how results may be described

- `free_running 1` on every daemon: offsets measured, clocks never steered (servo state s0 in all 504 radio-unit logs).
- Software timestamping only: microsecond noise against nanosecond targets. No absolute time-error claim is made.
- C1 was a port blackhole that removed *every* frame, a coarse version of selective interception.
- Raw PCAPs and runtime configs are not inside the campaign archive (configs are in a separate testbed archive).
- The harness scripts `randparams.py` and `scenarios.sh` changed after the original freeze. This is disclosed; the rules themselves still match their hashes.

Worklet status

MILESTONE	STATUS
M1 problem study, literature, architecture	DONE
M2 KPI and fault taxonomy (144 parameters × 16 faults)	DONE
M3 digital twin and fault injection	PARTIAL · injection done, twin not exercised
M4 AI model development	PARTIAL · evaluated, not usable on available telemetry
M5 healing logic integration	PARTIAL · decision engine evaluated, closed loop not validated
M6a testing and evaluation	DONE
M6b paper and IP	NOT STARTED

Not addressed: fault prediction; automated corrective action (only the 13 Sep pilot, 5/5 vs 0/5, with S15 not established); digital-twin validation; MTTR, recovery-time and availability figures.

Questions you will likely be asked

Q Is 0.990 a fair number if you revised the rule after seeing the data?

That is why both are shown. The pre-frozen base rule scored 0.625 on the same captures. v3 is a disclosed revision. v2, hashed at 11:41:31Z before the campaign started at 11:43:15Z, gives identical verdicts to v3 on all 168 runs, although v2 was developed after earlier same-day captures of the C-designs. An independently pre-registered re-run is listed as remaining work.

Q Is this real or simulated?

Real protocol software on a virtual wire. Six genuine linuxptp 4.0 daemons ran the actual election, cadence and servo, and attacks were real packets: a competing daemon, Scapy frames, tcpreplay, a port shut down. What is not real hardware: timestamps are software, all namespaces share one clock, and the clocks were not steered (free_running 1).

Q Who carries out these attacks?

Standards describe attackers by capability and position, not by name. ETSI TR 104 106 lists insiders, nation-states, cyber-criminals, hacktivists, cyber-terrorists and script kiddies. RFC 7384 classes them as internal or external, man-in-the-middle or injector. Six of our eight need an internal attacker on the fronthaul segment.

Q Did you make up the eight attacks?

No. Seven follow a published threat or test (RFC 7384 §3.2, ETSI T-SPLANE-01...04, ETSI TS 104 105 test IDs, TIMESAFE). C3 is our own instance of the RFC 7384 spoofing class.

Q What is the impact, and which module is hit?

The O-RU's timing (slave) port. A1: all 36 radio-unit logs elected the rogue as best master. A8: all 36 accepted the rogue BC. C1: all 36 lost their master on Announce timeout. In the literature, TIMESAFE observed an O-RU crash about 2 s after spoofing began. The service limits at stake are 3 μ s cell phase sync (TS 38.133) and 65 ns MIMO alignment (TS 38.104).

Q Which model gave you the KPIs?

None. The KPIs are read from PTP itself (packet fields and servo logs), as defined by IEEE 1588-2019 and G.8275.1. The ML model consumes six servo features; it does not produce KPIs.

Q Where is the dataset from?

We generated it on 20 Sep: 168 runs, 1,451,909 packet records, archive sha256 6149b4fb.... The only external data, TIMESAFE, is public and cited, and the August results that used it were withdrawn.

Q Does it work outside O-RAN?

The checks belong to PTP, which is used in power, broadcast, automotive and industry. So it is portable by design after re-deriving profile constants, but it has been demonstrated only on G.8275.1.

Q Why did ML do so badly? Did you just pick a weak model?

It had 6 of 28 planned features. The testbed has no GNSS, SyncE or oscillator data, and pmc returned nothing. Some attacks also starve its telemetry (88 servo samples per baseline run, 28 under C3). It also failed on attacks where telemetry was intact, so feature poverty and class overlap are the main causes. The fix is better telemetry, not a fancier classifier.

Q Where is the self-healing?

Not in this campaign. Verdicts map to recommended actions (isolate, holdover, escalate), but none was executed. A 13 Sep pilot executed a detector-triggered port failover (5/5 against 0/5 controls), and its independent validation was not established. Detection and classification are the prerequisite we have delivered.

Q Why is the boundary-clock replacement wrong every time?

The replacement identity is provisioned, but one clause of the base rule only checks the primary boundary-clock identity, and v3 cannot override a base-rule ATTACK by design. The cause is located and the fix is defined.

Q How do you know your data is not fabricated?

We audited our own pipeline and found two defects that had produced invalid data: copied replicates, and a congestion scenario that never injected. We fixed both, re-ran everything affected, withdrew the earlier 132-run campaign, and all 168 runs are hashed in the archive.

Words to know

Announce	PTP message in which a clock describes itself (identity, priority, clockClass, stepsRemoved) for the election.
BMCA	Best Master Clock Algorithm: the election that picks the master. G.8275.1 uses an alternate version.
Boundary clock (BC)	A clock that is a slave upstream and a master downstream, re-serving time.
clockClass	Quality grade of a clock: 6 locked to a primary reference, 7 holdover within spec, 248 default.
free_running 1	ptp4l setting: measure the offset but never adjust the clock.
G.8275.1	ITU-T telecom PTP profile with full timing support: domain 24–43, priority1 128, Sync 16/s, Announce 8/s.
Grandmaster (GM)	The top clock in the chain, usually locked to GNSS.
Holdover	Running on the local oscillator after losing the reference, while staying within spec for a while.
LLS-C1...C4	O-RAN's four ways of delivering time to the O-RU.
Macro average	Average of per-scenario rates, so each scenario counts equally.
McNemar test	Compares two classifiers on the same cases using only the cases where they disagree.
Namespace	An isolated Linux network stack; our "machines".
Offset	How far the slave's clock is from the master's.
pmc	linuxptp management client for querying data sets from a running ptp4l.
ptp4l	The linuxptp PTP daemon.
S-plane	The synchronization plane of the Open Fronthaul.
Sensitivity / specificity	Share of attacks called ATTACK / share of benign faults called BENIGN.
sequenceld	Per-message counter that must keep increasing; replay breaks it.
SHA-256 freeze	A fingerprint of a file taken before the experiment, proving it was not edited afterwards.
stepsRemoved	Number of clocks between this clock and the grandmaster.
tbf	Linux token-bucket filter, used to build the congestion bottleneck.
Wilson interval	A confidence interval for a proportion that behaves well at small n.

Sources for this guide: ORAN_SPlane_PRISM_Review_v8.pptx (text and speaker notes); verification_records/SESSION_RECORD_2026-10-03.md and V8_SLIDE_SOURCE_MAP.md; ORAN_SPlane_Packets_to_Classification_2026-09-29.xlsx (PER-RUN and worked-example sheets); 00_START_HERE.md. Standards cited are as listed on slide 42 of the deck. The two-machine hardware drift work of 2 Oct is outside the scope of this guide.